Robust One-Shot Bearing-Only Localization Front-End for Passive Multi-UAV Sensing

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Highlights

What are the main findings?

- The proposed corruption-aware front end reduces the strict catastrophic-failure rate at $0.2R$ from 0.293 to 0.040 in the outlier regime and from 0.273 to 0.068 in the mixed regime relative to least squares.
- Across 536 measured-data replay windows and 227 deadline-filtered replay windows, the method repeatedly contracts replay upper-tail error after stale or late bearings are discarded, even though stronger baselines still retain the best medians in some subsets.

What are the implications of the main findings?

- A passive multi-UAV sensing stack can use the method as a robust current-cycle front end to reduce catastrophic cue displacement before downstream tracking, handoff, or replanning.
- Fixed-budget screening is an Optional Engineering Extension for bandwidth- or latency-constrained cycles; all-sensor robust fusion remains the default when the full bearing set is available.

Abstract

Single-cycle bearing-only localization can fail catastrophically when a small subset of measurements is missing, biased, or grossly corrupted. This paper proposes a corruption-aware front end for passive multi-UAV sensing that combines consensus-style geometric seeding, trimmed iteratively reweighted Levenberg–Marquardt refinement, conditional common-bias correction, and an optional budget-constrained screening add-on. Across 3000 Monte Carlo cases, the strict catastrophic-failure rate at $0.2R$ drops from 0.293 to 0.040 in the outlier regime and from 0.273 to 0.068 in the mixed regime relative to least squares. Across 536 measured-data replay windows and 227 deadline-filtered replay windows, the method does not attain the best replay median but repeatedly contracts the 95th-percentile error tail, providing a reproducible front-end baseline for suppressing extreme cue failures in corrupted passive bearing fusion.

Keywords: bearing-only localization; passive target localization; robust estimation; budget-constrained screening; multi-UAV sensing; corrupted measurements

1. Introduction

Bearing-only passive sensing remains attractive for multi-UAV missions because it preserves low observability, reduces payload burden, and can still provide actionable target cues when active ranging or infrastructure support is unavailable [1–5]. In practice, however, the hardest failure mode is not lack of geometry alone. A one-shot localization cycle can fail because a few bearings are missing, heavily biased, grossly corrupted, or inconsistent with the rest of the set. Once those measurements enter the fusion step, the resulting position estimate may leave the local region in which it is still useful for downstream search, tracking, or handoff.

Recent UAV localization research has advanced quickly at the system level. Representative directions include tightly coupled integrated navigation, distributed cooperative localization, factor-graph or surveying-adjustment formulations, multi-condition constrained fusion, and learning-assisted cooperative inference [6–14]. A second line of work focuses on bearing-only sensing itself, covering AOA-based formation positioning, path optimization, observability enhancement, CRLB-driven trajectory design, and passive node selection [15–21]. These studies leave a current-cycle front-end problem open: given one fixed set of bearings, how should the estimator remain reliable when part of the set is corrupted?

This front-end problem is not trivial. Recent work on error bounds, dynamic bearing processing, dual-UAV filtering, bearing-only circumnavigation, underwater passive localization, and distributed passive sensing all show that corrupted angular information can dominate localization quality [22–31]. Yet the response is often to change the path, redesign the observer geometry, or invoke a heavier cooperative back end. Those are important directions, but they leave a practical gap at the front end of the sensing pipeline: a mission system often needs a robust answer now, before a new trajectory or graph optimization can be recomputed.

From a UAV-systems perspective, that front-end gap is concrete. In practical passive sensing, bearings may be produced by gimballed electro-optical payloads, direction-finding antennas, or compact angle-only perception stacks whose outputs are constrained by field of view, asynchronous triggering, varying pose quality, and limited uplink budget [32–35]. During aggressive turns or gusty low-altitude flight, gimbal vibration, transient non-line-of-sight blockage, and delayed inter-UAV packets can drive the angular error far away from the small-Gaussian assumption used by many clean geometric models. A planner or downstream tracker cannot benefit from those measurements if the localization front end collapses as soon as a few bearings become unreliable. This motivates a UAV-facing front-end question: how much robustness can be recovered inside one sensing cycle before invoking a heavier back end?

The second motivation is methodological. A large fraction of the localization literature either emphasizes geometry optimization or assumes a sequential estimator that can smooth errors over time. Those directions are important, but they do not remove the need for a robust one-shot estimate when a mission stack must react immediately. Moreover, sim-to-real studies in UAV autonomy repeatedly show that apparently small disturbances in attitude, delay, and synchronization can produce large downstream degradation if the front end is brittle [36,37]. For this reason, this study expands the validation chain beyond static Monte Carlo to include a measured-data multi-view replay surrogate with real visibility and asynchrony, a deadline-aware replay layer that discards stale or late bearings before fusion, pseudo-physical replay, and a multi-vehicle PyBullet replay layer.

This paper studies one-shot planar multi-observer bearing-only localization under corrupted measurements. The target is the fusion front-end inside a passive sensing stack: a robust estimator suppresses failures caused by bad bearings, and an optional screening stage is activated when only a limited number of measurements can be fused or transmitted.

This problem definition leads to three concrete design requirements. First, the front end must remain numerically stable when only a minority of bearings are unreliable, because a tactical UAV system often has to act before a new trajectory or full graph optimization is available. Second, the method should degrade gracefully when a bandwidth or compute budget forces the system to retain only part of the current bearing set; this motivates treating screening as a conditional budget-management module rather than as a universal replacement for full-set fusion. Third, the evidence should move beyond static geometry, because in practice the usefulness of a one-shot cue is determined by whether it remains consumable after timing variation, attitude disturbance, and navigation error are introduced. The architecture and validation chain used in the rest of the paper follow these three requirements directly.

The main contributions are threefold:

- a consensus-assisted robust bearing-only estimator that combines multi-start geometric initialization, trimmed iteratively reweighted refinement, optional common-bias correction, and a consensus fallback;
- an Optional Engineering Extension for fixed-budget screening, together with residual-only, reliability-only, geometry-only, and adaptive baselines that clarify when screening is useful and when all-sensor robust fusion should be kept;
- a reproducible validation chain comprising 3000 Monte Carlo cases across five corrupted-measurement regimes, a 2285-case fixed-budget screening benchmark, 536 measured-data multi-view replay windows with real visibility and synchronization, a 227-window deadline-aware measured-data replay layer with arrival filtering and staleness constraints, pseudo-physical dynamic replay with delay, jitter, per-sensor bias, and motion-induced disturbance, and a 1103-case PyBullet multi-vehicle replay study.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 formulates the corrupted-measurement localization problem. Section 4 presents the proposed estimation and screening framework. Section 5 describes the experimental protocol, Section 6 reports the results, and Sections 7 and 8 discuss the scope and conclude the paper.

2. Related Work

2.1. Bearing-Only and Passive Localization

Bearing-only and passive target localization remain active topics in UAV and distributed sensing research because they support low-emission observation and can be combined with cooperative formations [5,15–17]. Recent studies investigate D-criterion and Bayesian-FIM trajectory planning [18,21], CRLB-based variable-speed path design [19,33], observability enhancement with sensor-bias estimation [20], and bearing-only control with experimental disturbance handling [25,38]. Related developments also appear in underwater passive localization, wideband array localization, direct target localization, and passive source localization in other sensing domains [26–30,32,39]. The common thread is that geometry matters, but corrupted or uncertain bearings can still dominate the final error.

2.2. Robust Estimation Under Corrupted Angular Measurements

The literature increasingly recognizes that bearing quality is heterogeneous. Error-bound studies, dynamic bearing-angle analysis, multi-array likelihood processing, and dual-UAV filtering all show that uncertainty handling is at least as important as nominal geometry under realistic disturbances [22–24,40]. Classical least squares is usually too fragile in the presence of outliers, observer-pose uncertainty, or mixed corruption, while

generic stochastic search methods are often computationally heavy or hard to interpret. Standard robust choices such as Huber estimation [41], redescending M-estimation [42], RANSAC consensus filtering [43], damped Gauss–Newton / Levenberg–Marquardt updates [44,45], and graduated non-convexity [46] are therefore natural baselines, but they are rarely packaged as an explicit front-end block for fixed corrupted bearing sets in multi-UAV passive sensing. In particular, recent UAV papers still devote more attention to trajectory or formation design than to the behavior of the current-cycle estimator when a few bearings become severely unreliable [5,16,34].

2.3. Sensor Selection, Observability, and Budgeted Sensing

The second topic relevant to this paper is sensing selection. Passive node selection, observability-aware planning, and FIM/CRLB-based trajectory design are well established in the broader literature [17–21,33]. At the system level, collaborative UAV localization has also advanced through factor graphs, distributed inference, relative localization, and swarm-level cooperative frameworks [6,8–11,13,35]. The present work targets a specific front-end layer inside those broader frameworks. It addresses a centralized one-shot decision: once a set of bearings has already been produced for the current cycle, should all of them be fused, or should a smaller subset be retained because the current quality is too uneven? That framing is closer to front-end measurement triage than to long-horizon planning.

2.4. Validation Gap

Another gap is validation style. Public UAV-systems papers are increasingly expected to move beyond idealized static geometry and toward simulation pipelines that reflect timing, attitude, and coordination effects [36,37]. This is especially true for Drones-oriented work, where the distinction between a generic localization algorithm and a UAV-facing sensing paper often depends on whether the validation includes multi-vehicle trajectories, communication limits, or deployment-shaped disturbances. The current study still stops short of hardware-in-the-loop or onboard field tests, but it narrows that gap through measured-data replay, pseudo-physical replay, and PyBullet multi-vehicle replay while keeping the contribution centered on a corrupted-bearing front end.

Table 1 summarizes this positioning more compactly. The gap addressed here is not integrated navigation, distributed state estimation, or long-horizon geometry control in general. It is the current-cycle behavior of a bearing-only localization front end when the incoming angle set is partially corrupted and, in some cycles, budget-limited.

3. Problem Formulation

We consider one localization cycle with N observing platforms. The true planar observer positions are denoted by

$$\bar{s}_i = [\bar{x}_i, \bar{y}_i]^T, \qquad i = 1, \ldots, N, \tag{1}$$

and the target position is

$$p = [x, y]^T. \tag{2}$$

The estimator does not assume access to a full navigation stack. Instead, it receives reported observer poses

$$s_i = \bar{s}_i + n_{s,i}, \tag{3}$$

where $n_{s,i}$ models pose error from an external navigation module such as VIO, INS, or SLAM. This assumption resolves the usual tension between GNSS-denied mission language

Table 1. Study positioning against representative recent literature clusters.

Theme	Representative references	Primary emphasis	Gap relative to this work
Integrated navigation and cooperative localization	[6–10]	Integrated navigation and cooperative state estimation	Broader than current-cycle bearing triage; corrupted fusion is usually a secondary concern
Bearing-only geometry and observability design	[15,17–19,21]	Trajectory design, node selection, and geometry shaping	Focuses on how to acquire stronger bearings, not how to fuse a corrupted set that has already arrived
Dynamic passive localization and filtering	[22–25,29]	Sequential filtering and dynamic passive localization	Temporal estimation dominates; reusable one-shot front-end robustness is usually not the primary target
This work	–	Robust one-shot bearing fusion with optional budgeted screening	Targets outliers, mixed corruption, pose uncertainty, and heterogeneous bias in one front-end block

and known observer poses: the present paper studies the localization front end, not the upstream self-localization module.

Each valid observer contributes one bearing measurement:

$$\theta_i = \operatorname{atan2}(y - \bar{y}_i, x - \bar{x}_i) + b_c + b_i + \epsilon_i + o_i, \tag{4}$$

where b_c is a shared frame-level bias, b_i is a sensor-specific bias, ϵ_i is nominal noise, and o_i represents occasional gross corruption. A measurement may also be missing because of detection failure, occlusion, or communication loss. For a candidate target position p, the predicted bearing uses the reported pose:

$$\hat{\theta}_i(p; s_i) = \operatorname{atan2}(y - y_i, x - x_i). \tag{5}$$

The wrapped residual is

$$r_i(p, b_c) = \operatorname{wrap}\big(\hat{\theta}_i(p; s_i) - \theta_i - b_c\big), \tag{6}$$

where $\operatorname{wrap}(\cdot)$ maps an angle to $(-\pi, \pi]$.

The paper studies a one-shot planar setting with three explicit scope boundaries. First, the estimator is centralized and uses the current cycle only; it is not a sequential tracker or a distributed fusion protocol. Second, the reported observer poses are treated as inputs, but pose uncertainty is injected into the benchmark. Third, the geometry is planar; the paper targets fixed-height or small-height-difference settings and leaves full three-dimensional extension to future work.

The experimental thresholds are normalized by the formation radius R. In the main Monte Carlo benchmark, $R = 10$, so success at $0.1R$ corresponds to absolute error ≤ 1.0, and catastrophic failure at $0.5R$ corresponds to absolute error > 5.0. A stricter $0.2R$ boundary is reported separately as a tail-risk diagnostic rather than as a replacement for the main success/failure thresholds. Together, these thresholds distinguish locally usable cues from clearly misleading ones while still exposing heavy-tail behavior.

3.1. Operational Constraint Model

The manuscript treats measurement screening as an engineering response to limited fusion budget rather than as a universal replacement for all-sensor estimation. Let $z_i \in$

$\{0, 1\}$ indicate whether bearing i is transmitted or fused. Then the budgeted front-end problem is

$$\sum_{i=1}^{N} z_i \leq k, \qquad 3 \leq k < N, \tag{7}$$

where k reflects a fixed communication, computation, or timing budget for the current cycle. In practice, this budget corresponds to cases such as delayed uplink from a subset of vehicles, reduced onboard processing time, or a front-end module that is allowed to keep only a limited number of bearings before passing a candidate target cue downstream. A representative example is a 10-UAV round in which a TDMA uplink window can return only five or six bearing packets before the current fusion cycle expires, which is why the experiments use a budget close to 55% of the valid set. The baseline against which screening should be judged is therefore not "all-sensor robust fusion is always worse", but rather "when a subset must be used, can the subset be chosen more intelligently than with naive rules?".

3.2. Corruption Taxonomy and Applicability

The expanded benchmark distinguishes four practically important error sources. First, *outliers* model gross bearing corruption produced by spurious detections, local clutter, or momentary payload faults. Second, *missing observations* model occlusion, field-of-view truncation, or communication dropouts. Third, *shared or heterogeneous bias* models heading-reference mismatch, platform-frame misalignment, or per-sensor directional bias. Fourth, *observer-pose uncertainty* represents the fact that the localization front end depends on an upstream navigation subsystem whose errors propagate into the current localization cycle. These are still abstractions, but they map more cleanly to the UAV sensing stack than a single isotropic-noise assumption.

Figure 1 sketches the passive multi-UAV sensing scene considered in this study, whereas Figure 2 fixes the algorithmic boundary used throughout the experiments. Observer poses and bearing cues enter the corruption-aware front end, which outputs a single-cycle target cue for downstream tracking, handoff, or replanning.

3.3. Application Scenarios and User Pain Points

Three application scenarios motivate the current scope. The first is electromagnetically silent or low-emission cooperative observation, where several UAVs observe a noncooperative target without active ranging [5,16]. In this setting, the operator's pain point is not only global observability. The immediate problem is that a small number of bad bearings can move the fused cue far enough that a downstream confirmation camera or interceptor will search the wrong neighborhood. A front end that suppresses those large one-shot failures is therefore valuable even before one discusses long-horizon formation control.

The second scenario is communication-limited swarm cueing. In many practical stacks, not every vehicle can upload every measurement at full rate, especially when edge compute, bandwidth, or synchronization budget is limited. This is the use case behind the optional screening module. The question is not whether subset selection is universally superior to all-sensor fusion. The question is whether, once a subset must be retained, the system can avoid obviously poor triage decisions. Framed this way, the measurement-screening module addresses a concrete engineering bottleneck rather than an abstract optimization game.

The third scenario is degraded navigation and asynchronous payload operation. Passive bearings are often produced by vision, gimbal, or direction-finding modules whose timestamps, headings, and internal quality are not perfectly aligned. At the same time, the observer poses delivered by GNSS, VIO, INS, or SLAM are themselves imperfect. This

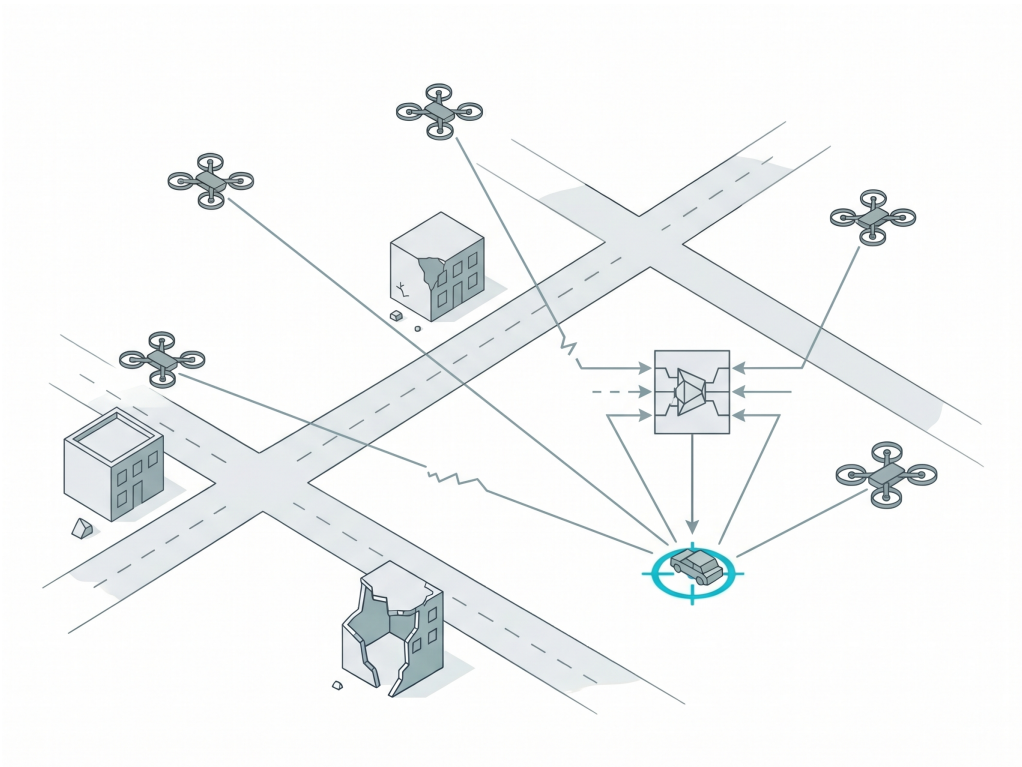

Figure 1. Passive multi-UAV bearing-only localization scenario considered in this paper. The illustration emphasizes partial occlusion, asynchronous reports, and corrupted line-of-bearing cues; quantitative assumptions are defined by the mathematical model and by Figure 2.

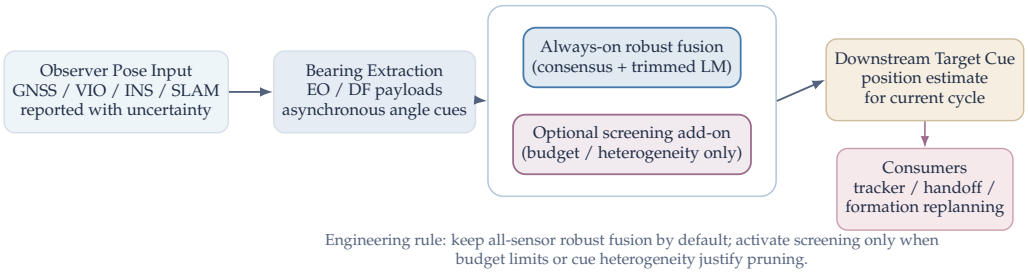

Figure 2. System position of the proposed method in a passive multi-UAV sensing stack. The paper focuses on the corruption-aware front end between upstream self-localization and downstream tracking, handoff, or replanning modules.

coupling between pose uncertainty and bearing corruption is precisely where plain least squares tends to become brittle. The present paper therefore focuses on the narrow question of what can be recovered inside one cycle before a tracker, planner, or human operator receives the cue.

Table 2 summarizes why the current corruption taxonomy is not merely an abstract noise menu. Each scenario maps to a concrete systems question that appears in passive UAV sensing: whether a cue remains usable under heading bias, whether limited uplink bandwidth should trigger subset triage, whether asynchronous payloads and pose drift destabilize the current estimate, and whether a simulation-only study is at least moving toward a more realistic validation chain. Making these correspondences explicit helps keep the manuscript focused on a front-end sensing problem with operational consequences rather than on a generic optimization exercise.

Table 2. Passive-sensing deployment scenarios and dominant corruption sources.

Scenario	Dominant corruption sources	Why the present front end is relevant
Electromagnetically silent formation sensing [5,16]	Shared heading bias, geometry imbalance, communication-limited updates	A robust one-shot fusion block suppresses catastrophic cue displacement before formation redesign or long-horizon planning is invoked.
Communication-limited swarm cueing [17,34]	Uneven node quality, delayed uplink, bandwidth budget	The fixed-budget screening module is explicitly motivated by this regime and is evaluated against residual-only, reliability-only, and geometry-only subset rules.
Degraded navigation and asynchronous payload operation [32,33]	Pose error, timestamp mismatch, intermittent detections, bearing drift	This failure pattern is matched directly by pose uncertainty, sensor bias, pseudo-physical replay, and PyBullet replay experiments.
Simulation-to-real transition studies [36,37]	Delay, attitude disturbance, bias drift, replay mismatch	The manuscript positions the method as an intermediate front-end baseline that should later be coupled to SITL, HITL, or field-log replay.

4. Materials and Methods

4.1. Framework Overview

The proposed framework has two stages. Stage 1 is a robust localization core. It receives all currently valid bearings, generates several candidate target hypotheses, refines them under trimmed iteratively reweighted estimation, and keeps the hypothesis with the smallest final residual. Stage 2 is optional. When only a limited number of bearings can be fused or transmitted, it screens the current set using geometry and residual reliability. An adaptive gate bypasses screening and keeps all-sensor robust fusion when the current bearing set looks sufficiently homogeneous.

This decomposition treats localization as an engineering front-end problem. The estimator sits between navigation/bearing extraction modules and downstream planning or tracking modules, and it is designed to remain lightweight enough for repeated use within the current sensing cycle.

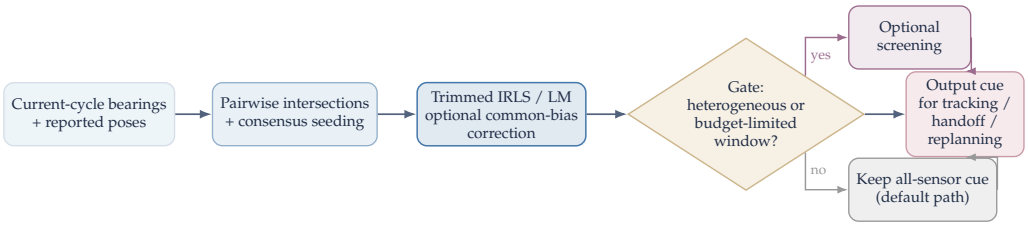

Figure 3. Front-end processing flow. Robust all-sensor fusion is the default path, while optional screening is activated only when communication limits or credibility dispersion justify subset pruning.

4.2. Consensus-Assisted Robust Estimation

Geometric candidate generation.

For observer i, the bearing ray is

$$\ell_i(\tau_i) = s_i + \tau_i \begin{bmatrix} \cos\theta_i \\ \sin\theta_i \end{bmatrix}. \tag{8}$$

Pairwise ray intersections generate an initial candidate pool. The implementation keeps several consensus-style representatives of that pool, including the centroid, the coordinate-wise median, and a trimmed centroid. In addition, a sampled consensus prior is extracted so that heavily contaminated cases have an explicit fallback hypothesis before iterative refinement begins.

Robust objective and Jacobian.

For active measurements, the estimator minimizes

$$\min_{p,b_c} \sum_{i=1}^{N} w_i \, \rho(r_i(p, b_c)), \tag{9}$$

where $\rho(\cdot)$ is implemented through iteratively reweighted least squares and the weights are updated after trimming large residuals. Let

$$d_i^2 = (x - x_i)^2 + (y - y_i)^2. \tag{10}$$

The bearing Jacobian with respect to the target position is

$$\mathbf{J}_i(p) = \begin{bmatrix} -\dfrac{y - y_i}{d_i^2} & \dfrac{x - x_i}{d_i^2} \end{bmatrix}. \tag{11}$$

The code uses a damped Gauss–Newton/Levenberg–Marquardt step rather than finite-difference gradient descent. With $A_i = [\mathbf{J}_i \quad -1]$ when common-bias correction is active, the linearized update solves

$$\Delta = -\left(A^T W A + \lambda I\right)^{-1} A^T W r, \tag{12}$$

where W is the diagonal matrix of current robust weights and λ is a small damping term.

Trimmed iteratively reweighted refinement.

The robust core updates weights from the residual pattern, trims the largest residuals, and then refines the estimate by several damped Gauss–Newton steps. Redescending down-weighting is used after trimming so that the tail of the residual distribution is suppressed more aggressively than in plain Huber IRLS. The common-bias term is updated only when the active residuals are sufficiently concentrated; otherwise the algorithm falls back to zero bias rather than forcing an unstable common correction onto heterogeneous measurements. After refinement, a consensus fallback is retained when the residual tail remains too heavy or when the trimmed solution drifts away from the consensus hypothesis without a clear residual advantage. In this sense, the proposed estimator is a compact robust front end that combines geometric hypotheses, trimming, reweighting, and consensus rescue.

Algorithmic summary.

The robust estimator can be summarized as follows. First, generate a pool of pairwise-intersection hypotheses together with consensus-style summary points. Second, evaluate a sampled consensus fallback so that a heavily corrupted case still has a viable initial anchor. Third, refine each candidate by several rounds of trimmed reweighting and damped Gauss–Newton updates, optionally correcting a shared bias when the active residuals are concentrated enough. Fourth, retain the candidate with the lowest final robust residual. Fifth, if the residual tail after refinement remains too heavy, fall back to the best consensus hypothesis. This sequence remains simple enough to reproduce from the codebase while still addressing the main failure modes seen in corrupted bearing sets.

Table 3. Internal Stage-1 decision rules used by Algorithm 1. Here $\delta_H = 0.08$ denotes the fixed Huber scale used throughout the reported experiments.

Rule	Quantitative trigger in the implementation	Purpose
Coherence shortcut	$\mathrm{median}(\lvert r_{\mathrm{LS}}\rvert) \leq 0.58\delta_H$, $\mathrm{MAD}(\lvert r_{\mathrm{LS}}\rvert) \leq 0.32\delta_H$, $\mathrm{P90}(\lvert r_{\mathrm{LS}}\rvert) \leq 1.65\delta_H$, and $\mathrm{mean}(\lvert r_{\mathrm{LS}}\rvert) \leq 0.95\delta_H$	Accept a low-cost smooth hypothesis directly when the current window already looks internally consistent.
Bias activation	At least three active bearings and $\mathrm{std}(r_{\mathrm{active}}) \leq 1.8\delta_H$; otherwise the shared bias is forced to zero	Avoid fitting an unstable common bias when residuals are too dispersed.
Severe residual flag	Trim $\lfloor 0.15N \rfloor$ largest residuals, then additionally flag any $\lvert r_i\rvert > \max(1.75\delta_H, Q_{0.78}(\lvert r\rvert))$, while always keeping at least three active bearings	Convert vague "large residual" wording into a reproducible rule for aggressive outlier suppression.
Consensus rescue	Return the consensus candidate if the refined severe-residual fraction satisfies $\mathbb{E}[\mathbb{I}\{\lvert r\rvert > 1.75\delta_H\}] \geq 0.12$, or the removed count is at least $\max(1, 0.20N)$, or the refined point is more than 1.0 away from the consensus point while the consensus residual is within $+0.03$ and either the removed count is at least $\max(2, 0.25N)$ or $\lvert b\rvert \geq 0.08$	Recover from heavy-tail windows in which the refined branch becomes suspicious despite a low nominal residual.
Smooth rescue	Return the best smooth hypothesis if the refined branch removes at least $\max(2, 0.22N)$ bearings, drifts more than 0.65 from the smooth point, and fails to improve the residual by more than 0.010; also rescue the smooth branch whenever the refined residual is within $+0.015$ of the smooth residual	Replace ambiguous phrases such as "prunes too aggressively" and "not clearly better" with explicit thresholds.

Algorithm 1 Consensus-Assisted Robust Fusion

Require: Reported observer poses $\{s_i\}_{i=1}^N$, current valid bearings $\{\theta_i\}_{i=1}^N$
Ensure: Current-cycle target cue $\hat{p}$
Generate pairwise-intersection hypotheses and consensus summary points
Form a small set of smooth hypotheses and one consensus-style fallback seed
if the coherence shortcut in Table 3 is satisfied **then**
  Return the smooth candidate with the lowest residual
end if
for all geometric candidates and the consensus candidate **do**
  Activate shared-bias correction only if the bias rule in Table 3 is satisfied
  Trim large residuals and update redescending weights
  Refine the candidate by damped Gauss–Newton / LM steps
end for
Keep the candidate with the lowest final robust residual
if the consensus-rescue rule in Table 3 is triggered **then**
  Return the consensus hypothesis
else if the smooth-rescue rule in Table 3 is triggered **then**
  Return the smooth hypothesis
else
  Return the refined robust candidate
end if

4.3. Reliability-Adaptive Measurement Screening

The screening stage starts from the pilot estimate delivered by Stage 1. Residual reliability is computed from the current residuals, and screening is invoked only in fixed-budget settings or when the adaptive gate considers the current residual pattern too heterogeneous to trust all measurements equally.

For a candidate subset $\mathcal{S}$, the weighted geometry matrix is

$$\mathbf{F}_{\mathcal{S}} = \sum_{i \in \mathcal{S}} w_i \mathbf{J}_i^T \mathbf{J}_i, \tag{13}$$

and the isotropy term is

$$\eta_{\mathcal{S}} = \frac{\lambda_{\min}(\mathbf{F}_{\mathcal{S}})}{\lambda_{\max}(\mathbf{F}_{\mathcal{S}})}. \tag{14}$$

The proposed normalized screening score is

$$\mathrm{Score}(\mathcal{S}) = 0.34\,\phi_{\det}(\mathcal{S}) + 0.18\,\eta_{\mathcal{S}} + 0.24\,\bar{w}_{\mathcal{S}} + 0.24\,\phi_r(\mathcal{S}), \tag{15}$$

where

$$\phi_{\det}(\mathcal{S}) = 1 - \exp\Big(-10^4 \det(\mathbf{F}_{\mathcal{S}})\Big), \qquad \phi_r(\mathcal{S}) = 1 - \min\Big(\frac{\bar{r}_{\mathcal{S}}}{0.25}, 1\Big), \tag{16}$$

and $\bar{w}_{\mathcal{S}}$ and $\bar{r}_{\mathcal{S}}$ denote the mean reliability and mean absolute residual on the subset. The score uses four normalized ingredients and avoids stacking several strongly correlated geometry terms.

The two geometry terms can be interpreted as local FIM surrogates. Specifically, $\phi_{\det}$ plays the role of a normalized D-optimality proxy and $\eta_{\mathcal{S}}$ behaves like a normalized E-optimality or isotropy proxy, both of which are commonly used in observability-aware trajectory and node-selection studies [18,19,21]. The remaining two terms inject current-cycle credibility information that those geometry-only criteria do not capture.

The score approximates a small budgeted search problem. In idealized form, the screening objective can be expressed as

$$\mathcal{S}^\star = \arg\max_{|\mathcal{S}|=k} \Big[\alpha_1 \phi_{\det}(\mathcal{S}) + \alpha_2 \eta_{\mathcal{S}} + \alpha_3 \bar{w}_{\mathcal{S}} + \alpha_4 \phi_r(\mathcal{S})\Big], \tag{17}$$

with fixed coefficients $\alpha_1, \ldots, \alpha_4$ chosen once and then held constant across the reported experiments. The implementation evaluates this objective over a capped reliability-sorted pool of size $m = \min(N, \max(k+2, 8))$ so that the search remains computationally small while still exploring more than the final subset size.

The proposed policy performs a small exhaustive search over a top-reliability candidate pool, which is feasible because the pool is capped in the experiments. Baselines include random subset selection, angular-spread greedy selection, FIM-only greedy selection, residual-only top-k, reliability-only top-k, and the all-sensor robust estimator. The adaptive policy uses all-sensor robust fusion when the residual median absolute deviation and reliability dispersion are both small; otherwise it activates screening. This makes screening a conditional tool rather than a default law. Section 6 therefore reports both score-term ablation and explicit coefficient perturbation so that the selector is not justified by one frozen coefficient vector alone.

Algorithm 2 Adaptive Budget Screening

Require: Pilot estimate $\hat{p}$, residuals $\{r_i\}$, reliabilities $\{w_i\}$, current budget k
Ensure: Subset $\mathcal{S}$ or the all-sensor decision
  Compute residual dispersion and reliability dispersion on the current window
  if the window is homogeneous and the full set can be fused **then**
    Keep all current valid bearings
  else
    Form a capped candidate pool by reliability ranking
    Enumerate budget-feasible subsets in that pool
    Score each subset with normalized geometry, isotropy, reliability, and residual terms
    Return the highest-scoring subset
  end if

Cue-usability metadata.

Besides the coordinate cue $\hat{p}$, the front end exposes three lightweight diagnostics to downstream modules: the residual median absolute deviation, the local isotropy proxy η, and the active-bearing fraction after trimming. In deployment, these quantities can be combined into a simple usability score or passed directly to a tracker for gating and down-weighting. These metadata are engineering diagnostics rather than calibrated probabilities, and they help a downstream consumer distinguish a numerically available cue from one that is geometrically weak or residual-inconsistent.

4.4. Complexity and Reproducibility

The geometric candidate generation stage is $O(N^2)$ because it depends on pairwise intersections. Each inner Gauss–Newton update is linear in the number of active measurements, while the number of outer reweighting rounds is fixed by configuration. In the screening stage, the exhaustive search is confined to a small top-reliability pool, so the practical cost is bounded by $\binom{m}{k}$ with small m rather than by exhaustive search over the full sensor set. Error amplification follows the same geometry logic: when the effective information matrix becomes ill-conditioned, small angular perturbations can expand into large position errors roughly in proportion to $1/\lambda_{\min}(\mathbf{F})$. This is why the front end monitors both residual dispersion and isotropy instead of trusting geometry or residual size alone. All experiments reuse the same Python package interfaces, and all manuscript figures are generated from frozen JSON outputs rather than from manually edited spreadsheets.

4.5. Implementation Details and Default Settings

The implementation uses a trim ratio of 0.15, three outer reweighting rounds, Huber scale 0.08, Tukey cutoff 0.18, and LM damping 10^{-3} as default settings. The adaptive gate

keeps all-sensor fusion when the residual median absolute deviation is below 0.05 and the reliability dispersion is below 0.12. The four screening coefficients are fixed after a small validation sweep and then held constant as a tractable surrogate for the budgeted geometry–credibility trade-off studied here. The fixed-budget screening benchmark sets the budget to approximately 55% of currently valid bearings, which mimics a short TDMA-style uplink window in which only five or six out of ten observers can forward cues within one localization cycle. The exhaustive search is confined to a capped top-reliability pool. These settings are not retuned from regime to regime; they are fixed across the expanded experimental matrix. Appendix L lists the exact values together with their engineering interpretation and the experiments in which they were stress-tested.

During the preparation of this manuscript, ChatGPT was used exclusively for language polishing. All technical methodologies, numerical claims, and results were independently verified by the authors.

5. Experimental Protocol

5.1. Main Monte Carlo Benchmark

The main synthetic benchmark contains 3000 cases. Five corrupted-measurement regimes are evaluated: clean, outlier, mixed, pose uncertainty, and heterogeneous bias. Each regime is run over 200 random seeds and three formation families (circular, random, and near-degenerate narrow-arc layouts) with eight observers. The formation radius is fixed at $R = 10$, but the target is randomized reproducibly inside the formation footprint rather than fixed at one convenient coordinate. This removes dependence on one nominal geometry while preserving exact reproducibility. If a practical swarm operates with a 100 m observation radius, the normalized thresholds used in this paper map to 10 m at $0.1R$ and 50 m at $0.5R$. The outlier regime uses outlier rate 0.28 and scale 0.42. The mixed regime combines common bias 0.04, missing rate 0.20, outlier rate 0.22, and outlier scale 0.45. The pose-uncertainty regime injects observer-pose noise of standard deviation 0.35. The heterogeneous-bias regime uses sensor-specific bias standard deviation 0.05 in addition to a small common bias.

The baseline estimators are nonlinear least squares, Huber IRLS, GNC with a Geman–McClure robustifier, and RANSAC. The proposed method is evaluated under the same interface and with the same reported observer poses. The main metrics are median error, mean error, 90th-percentile error, success at $0.1R$, and catastrophic failure above $0.5R$. A stricter $0.2R$ catastrophic threshold is reported separately as a supplemental heavy-tail metric. Bootstrap 95% confidence intervals are reported for the medians.

5.2. Fixed-Budget Screening Benchmark

The screening benchmark contains 2285 valid fixed-budget cases over mixed and severe regimes, four formation families (circle, perturbed, random, and near-degenerate), and 8/10/12-observer settings. Cases with fewer than four valid bearings after corruption are discarded because fixed-budget subset selection is not meaningful there. The subset budget is approximately 55% of the currently valid measurements, clipped to remain strictly smaller than the full set. In concrete terms, this corresponds to retaining roughly five or six bearings out of a ten-vehicle observation round when uplink, onboard compute, or fusion latency forces partial transmission. The benchmark tests whether a budgeted policy can use the limited subset more effectively than simple residual-only, reliability-only, geometry-only, or random screening.

5.3. Pseudo-Physical Dynamic Replay

To narrow the gap between static geometry and UAV-like dynamics, a pseudo-physical replay benchmark is also included. The replay generator injects motion-dependent delay, per-sensor bias, attitude-jitter-like disturbance, missing observations, and outliers before the current fusion cycle is evaluated. Forty seeds, three formations, and three swarm sizes produce 720 dynamic cases over mild and harsh regimes. This layer remains simulation-based, but it is closer to flight-like corruption than a purely static Monte Carlo sweep.

5.4. Public Real-Flight Trajectory Replay

To strengthen the validation chain with a measured data source, a measured-data multi-view replay surrogate is added using dataset 3 from the `drone-tracking-datasets` collection [47]. The dataset provides surveyed observer locations, manual multi-camera detections, cross-camera synchronization parameters, and an RTK target trajectory. Rather than treating those detections as direct onboard bearing measurements, the replay layer uses the measured trajectory, real visibility masks, and synchronization tables to rebuild current-cycle angle cues under real multi-view geometry. Concretely, the real inputs in this layer are the surveyed observer poses, measured visibility masks, cross-camera synchronization tables, and RTK target trajectory supplied by the dataset. Each replayed bearing is then reconstructed from the synchronized observer–target geometry by projecting the RTK target position into the observer frame and taking the azimuth implied by the synchronized time stamp, so the replay inherits measured geometry and measured asynchrony but not the original image-space detector noise. The reported outlier, bias, and pose perturbations are injected only after this reconstruction step, which keeps the stress-test corruption controlled and reproducible instead of conflating it with the dataset's raw annotation process. Camera 0 is used as the reference stream, 450 reference windows are sampled over the full sequence, and the current cycle is retained only when at least three observers remain visible after synchronization and visibility filtering.

Two replay regimes are used. The nominal regime adds mild pose uncertainty, common bias, per-sensor bias, and a low outlier rate. The disturbed regime increases pose uncertainty, timing jitter, and outlier severity. This produces 536 retained replay windows in total. The value of this layer is specific: it is not a field trial of the full UAV system, but it is substantially closer to measured target motion, real observer geometry, and real asynchronous visibility than the purely synthetic layers.

5.5. Deadline-Aware Measured-Data Replay

To make the measured-data replay layer more system-facing, a second pass is added in which only bearings that reach the current fusion cycle on time are allowed into the estimator. Starting from the same public replay windows, each valid bearing inherits a synchronization age from the measured cross-camera frame mapping and then receives sampled link and queue latency. Bearings are discarded if they are lost in transit, exceed a staleness budget, or arrive after the front-end deadline minus a small processing guard. The nominal deadline regime uses a 170 ms fusion deadline with a 28 ms guard, leaving 142 ms of usable budget; the disturbed regime uses 300 ms and 40 ms, leaving 260 ms. Across the original 536 measured-data windows, this filter retains 227 windows in total, with 152/257 nominal windows and 75/279 disturbed windows surviving the deadline gate. The corresponding mean on-time bearing counts are 2.78 and 1.97 per original cycle, while every retained window still satisfies the estimator's minimum three-bearing requirement. This layer tests a concrete systems question: how robust is the front end once the current cycle can only fuse the bearings that are both visible and on time?

5.6. PyBullet Multi-Vehicle Replay

To further reduce the gap between static geometry and UAV-like temporal disturbance, we also replay bearing measurements from a multi-vehicle PyBullet environment derived from the `gym-pybullet-drones` ecosystem [36]. The replay uses circle and ellipse formations with 8 and 10 vehicles, warm-up phases, time-sampled trajectories, gust-like disturbance, delayed observer poses, attitude-induced bearing perturbation, yaw-defined sensor field of view with a payload mounting offset, soft field-of-view dropouts, missing observations, and slowly drifting common bias. The resulting benchmark contains replayed localization cycles over nominal and disturbed regimes. Although the current study still does not reach hardware-in-the-loop, this replay layer now serves as a dynamic synthetic replay layer that complements the measured-data surrogate [37].

5.7. Additional Generalization and Efficiency Analyses

Beyond the headline benchmarks, the manuscript includes five auxiliary analyses. First, a formation-generalization study compares circular, elliptical, perturbed, and random observer layouts under mixed corruption. Second, sensor-count scaling evaluates 4/6/8/10/12-observer settings for both circle and random formations. Third, sensitivity sweeps vary outlier rate, common bias magnitude, and nominal noise level. Fourth, ablation, observability, and runtime studies isolate the effect of trimming, bias estimation, reweighting, geometry quality, and computational burden. Fifth, two selector diagnostics are reported in the main text: score-term ablation and randomized mild/moderate/strong coefficient perturbations for the normalized screening score. These analyses do not replace the main benchmark, but they are important for showing where the method is fragile, where it generalizes, and what computational price is paid for the robustness gains.

6. Results

6.1. Robust Localization Under Corrupted Bearings

Table 4 and Figure 4 report the main result: corrupted measurements cause clear failures for least squares, while robust front-end estimation materially reduces both median error and failure risk. In the outlier regime, the median error decreases from 1.1186 for least squares to 0.2108 for the proposed method. In the mixed regime, it decreases from 1.0015 to 0.4346. These are not marginal differences: they represent a shift from brittle one-shot localization to a front-end that stays usable after bad bearings enter the set.

Table 4. Main synthetic benchmark across five corruption regimes and three formation families.

Regime	LS	Huber	GNC-GM	RANSAC	Proposed
Clean	0.1200	0.1200	0.1201	0.1200	0.1201
Outlier	1.1186	0.4492	0.4471	0.2091	0.2108
Mixed	1.0015	0.5219	0.5039	0.4315	0.4346
Pose Uncertainty	0.6985	0.4898	0.4899	0.4526	0.4562
Heterogeneous Bias	0.5303	0.4244	0.4223	0.4148	0.4212

The comparison against stronger robust baselines is also informative. Huber IRLS and GNC-GM are both much better than least squares, but they remain consistently behind the proposed front end in the mixed, pose-uncertainty, and heterogeneous-bias regimes. RANSAC is the strongest competing baseline in the current benchmark: the proposed estimator often matches it in median error across the hard regimes and exceeds it mainly on tail-oriented metrics rather than through universal center-error superiority. This pattern is consistent with the design of the method: consensus remains crucial under gross corruption, while the trimmed reweighting and bias-aware fallback keep one unified front-end pipeline usable when the corruption mode changes from one window to the next.

Table 5. Detailed statistics in the two hardest synthetic regimes. Thresholds are normalized by the formation radius $R = 10$.

Regime	Method	Median [95% CI]	Success@0.1R	Cat.@0.5R
Outlier	Least Squares	1.1186 [1.0158, 1.2892]	0.4533	0.0733
Outlier	GNC-GM	0.4471 [0.3987, 0.4769]	0.7883	0.0333
Outlier	RANSAC	0.2091 [0.1937, 0.2373]	0.9067	0.0050
Outlier	Proposed	0.2108 [0.1988, 0.2456]	0.9050	0.0050
Mixed	Least Squares	1.0015 [0.9029, 1.1219]	0.4983	0.0633
Mixed	GNC-GM	0.5039 [0.4641, 0.5533]	0.7583	0.0450
Mixed	RANSAC	0.4315 [0.4004, 0.4556]	0.8550	0.0167
Mixed	Proposed	0.4346 [0.4017, 0.4645]	0.8517	0.0200

The pose-uncertainty and heterogeneous-bias results are particularly important for the scope of the paper. They show that the expanded benchmark is no longer built around ideal observer poses and shared-noise assumptions only. Under pose uncertainty, the median error decreases from 0.6985 for least squares to 0.4562 for the proposed method. Under heterogeneous bias, it decreases from 0.5303 to 0.4212. These improvements are smaller than in the outlier regime, but they address more realistic failure modes for practical passive sensing.

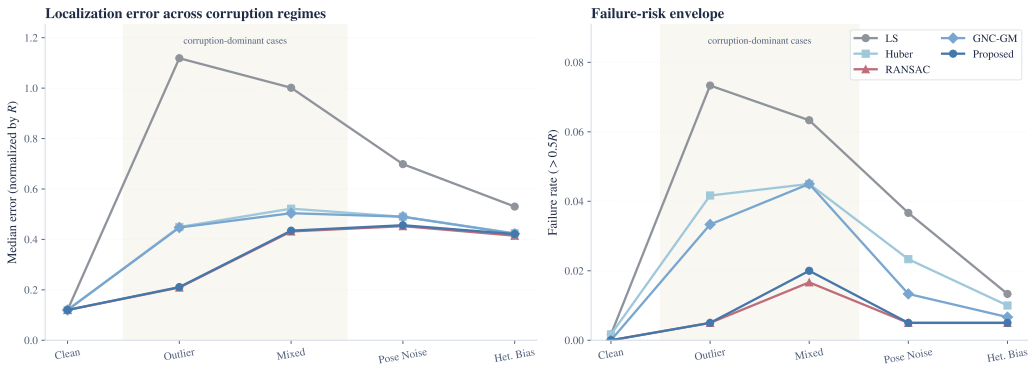

Figure 4. Main benchmark across five corruption regimes. Left: median localization error. Right: catastrophic-failure rate. The proposed estimator is designed for failure-risk suppression rather than universal dominance.

Figure 5 adds a distribution-level view for the outlier-rich and mixed-corruption regimes. In the outlier-rich regime, the proposed estimator shifts the full error distribution leftward relative to least squares and GNC-GM while nearly overlapping RANSAC over most of the range. The 90th-percentile error drops from 3.8429 for least squares to 0.9506 for the proposed method, and the success rate at $0.1R$ increases from 0.4533 to 0.9050. In the mixed-corruption regime, the same pattern persists: the 90th percentile drops from 3.8537 to 1.3820 and the success rate increases from 0.4983 to 0.8517. The improvement therefore extends beyond a small set of catastrophic trials and appears across the empirical distribution, which is what a robust front end should achieve before a downstream tracker or planner is invoked.

Figure 6 and Table 6 make the same point under stricter failure thresholds and paired tests. The main paper still uses $0.1R$ and $0.5R$ as its standard cue-quality boundaries; the $0.2R$ line is added only to expose heavy-tail sensitivity. Under that stricter diagnostic, the outlier-regime failure rate drops from 0.2933 for least squares to 0.0400 for the proposed estimator, and the mixed-regime failure rate drops from 0.2733 to 0.0683. The paired Wilcoxon tests and Cliff's δ show practically meaningful gains over least squares in both hard regimes, with medium-to-large effect sizes for the strongest tail-risk contrasts. Against GNC-GM, the effect is medium in the outlier regime and small in the mixed regime; against RANSAC, the effect is negligible, which is consistent with RANSAC already being strong

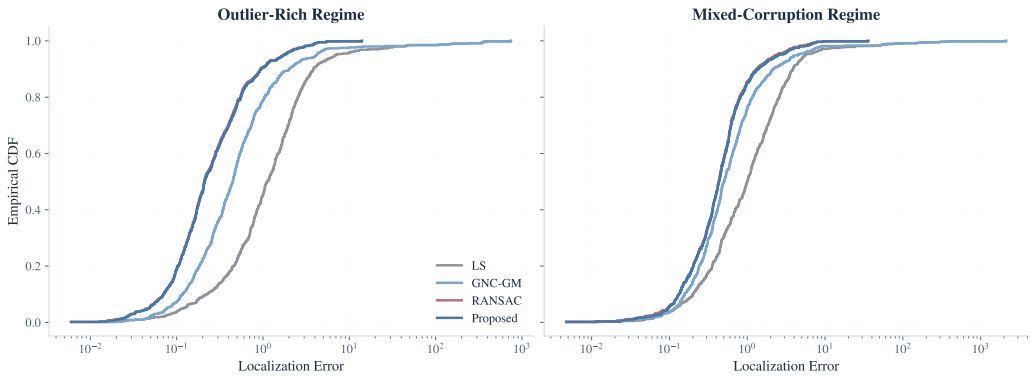

Figure 5. Empirical CDFs in the outlier-rich and mixed-corruption regimes. The proposed estimator stays close to the strongest robust baseline while shifting the distribution left of least squares and GNC-GM, especially in the upper tail.

in gross-outlier conditions. The main value of the proposed front end is therefore not universal center-error superiority but a single robust pipeline that remains usable when bias, missing data, and pose uncertainty appear together.

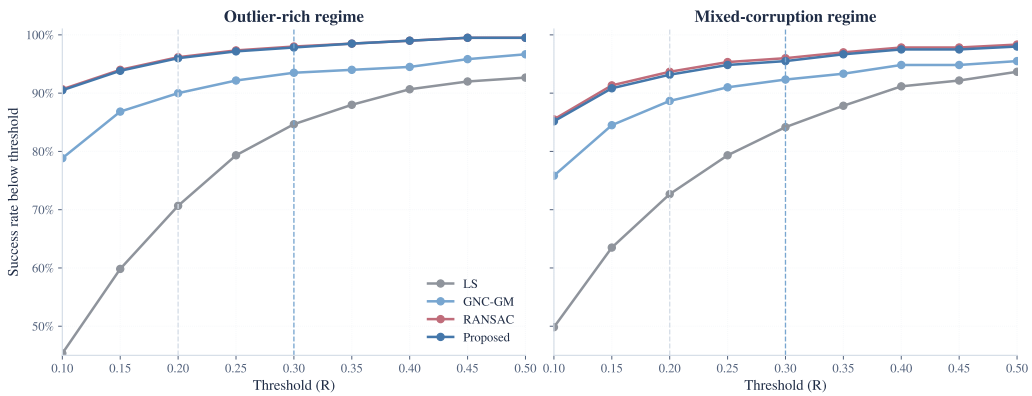

Figure 6. Success-rate sweep under stricter error thresholds. The proposed front end preserves a large margin over least squares as the acceptable cue radius contracts from $0.5R$ to $0.1R$.

Table 6. Paired tail-risk, significance, and effect-size summary in the two most demanding synthetic regimes. Positive Δ indicates lower error for the proposed front-end; positive Cliff's δ means the baseline error is larger than the proposed error more often than the reverse.

Regime	Comparison	Wins–Losses–Ties	Wilcoxon p	Median Δ	Cliff's δ	P95 Baseline / Proposed	P99 Baseline / Proposed
Outlier	Proposed vs LS	521–70–9	2.35×10^{-77}	0.7637	0.656 (large)	6.9637 / 1.7526	159.7861 / 3.8781
Outlier	Proposed vs GNC-GM	385–134–81	6.16×10^{-33}	0.0953	0.333 (medium)	4.1850 / 1.7526	150.1139 / 3.8781
Outlier	Proposed vs RANSAC	93–86–421	0.0357	0.0000	-0.007 (negligible)	1.7049 / 1.7526	3.8781 / 3.8781
Mixed	Proposed vs LS	411–169–20	2.39×10^{-42}	0.3258	0.403 (medium)	5.5785 / 2.5071	76.1232 / 6.7965
Mixed	Proposed vs GNC-GM	293–161–146	2.32×10^{-14}	0.0000	0.133 (negligible)	4.5027 / 2.5071	62.6256 / 6.7965
Mixed	Proposed vs RANSAC	78–107–415	0.9992	0.0000	-0.007 (negligible)	2.3245 / 2.5071	6.4788 / 6.7965

6.2. Why Not Just RANSAC?

RANSAC is the right baseline to confront directly because it is already very effective in pure gross-outlier regimes. The results above show that the proposed estimator does not materially outperform RANSAC in the outlier benchmark, and that is an important boundary condition for the paper. The motivation for keeping the proposed front end is different: the same pipeline must also remain stable when heterogeneous error mechanisms interact with gross corruption. In those regimes, the proposed estimator remains comparable with RANSAC in median accuracy while preserving one consistent refinement-and-fallback stack for the rest of the manuscript.

Algorithm 1 uses an internal sampled-consensus prior inside Stage 1, but the external comparison remains non-circular. The internal prior is only one candidate inside a broader

estimator that also evaluates smooth hypotheses, trimmed Gauss–Newton refinement, optional bias correction, and fallback logic. The external baselines are executed independently as standalone competitors with their own objectives and stopping rules. The manuscript therefore compares complete front ends against complete baseline estimators while also making clear that consensus itself remains an important ingredient.

Table 7 decomposes that point in the mixed-corruption regime. Pure RANSAC has median error 0.5241 and P95 error 2.4951. Adding LM polishing to the consensus inliers captures most of the median improvement, reducing the median to 0.4233 and P95 to 2.2843. The complete proposed pipeline is close to the polished RANSAC branch in center accuracy, with median 0.4267, but gives a slightly lower mean, P95, and $0.2R$ catastrophic-failure rate over the same 717 retained cases. This ablation therefore supports a bounded claim: the full pipeline improves substantially over an unpolished consensus hypothesis and is at least competitive with a simple RANSAC+LM repair under mixed corruption, while the strongest standalone RANSAC baseline remains an important reference rather than a defeated strawman.

Table 7. Incremental RANSAC-family ablation in the mixed-corruption regime. The table separates the sampled consensus hypothesis, LM polishing on the consensus inliers, and the complete proposed front end.

Method	Median	Mean	P90	P95	Cat. $> 0.2R$
Pure RANSAC	0.5241	0.8195	1.4238	2.4951	0.0628
RANSAC + LM	0.4233	0.7665	1.2817	2.2843	0.0628
Proposed full pipeline	0.4267	0.7490	1.2889	2.2740	0.0614

Figure 7 expands the old single-window illustration into a four-case gallery aligned with likely reviewer questions. The first row shows a pure gross-outlier window in which RANSAC is the strongest method, confirming that the paper is not trying to erase RANSAC's value. The second row shows mixed corruption, where trimmed robust fusion becomes the best compromise. The third row shows pose uncertainty with difficult geometry, where consensus alone is less stable. The fourth row is a heterogeneous-bias limitation case in which the proposed method no longer dominates the strongest competitor. This gallery makes the scope boundary visual rather than rhetorical.

6.3. Fixed-Budget Screening Is Conditional, Not Universal

The screening benchmark confirms a conditional story. All-sensor robust fusion remains the unconstrained upper bound, with overall median error 0.4981. Under fixed subset budgets, however, screening is still useful. The proposed geometry–reliability policy improves the overall median error from 0.6118 for residual-only screening and 0.6383 for reliability-only screening to 0.5872. The adaptive variant reduces the overall median further to 0.5518 and improves the severe-regime median from 0.7774 to 0.7487 (Table 8).

Table 8. Fixed-budget screening benchmark. All-sensor robust fusion remains the unconstrained upper bound, while adaptive and geometry–reliability screening improve over simpler subset rules under fixed budgets.

Policy	Overall Median	Mixed Median	Severe Median	Success@1.0	Cat.@5.0
All-Sensor Robust	0.4981	0.3404	0.7022	0.8166	0.0306
Residual-Only	0.6118	0.4497	0.7913	0.8096	0.0302
Reliability-Only	0.6383	0.4970	0.8145	0.7777	0.0319
Geom+Reliability	0.5872	0.4300	0.7774	0.7996	0.0315
Adaptive	0.5518	0.3857	0.7487	0.8026	0.0311

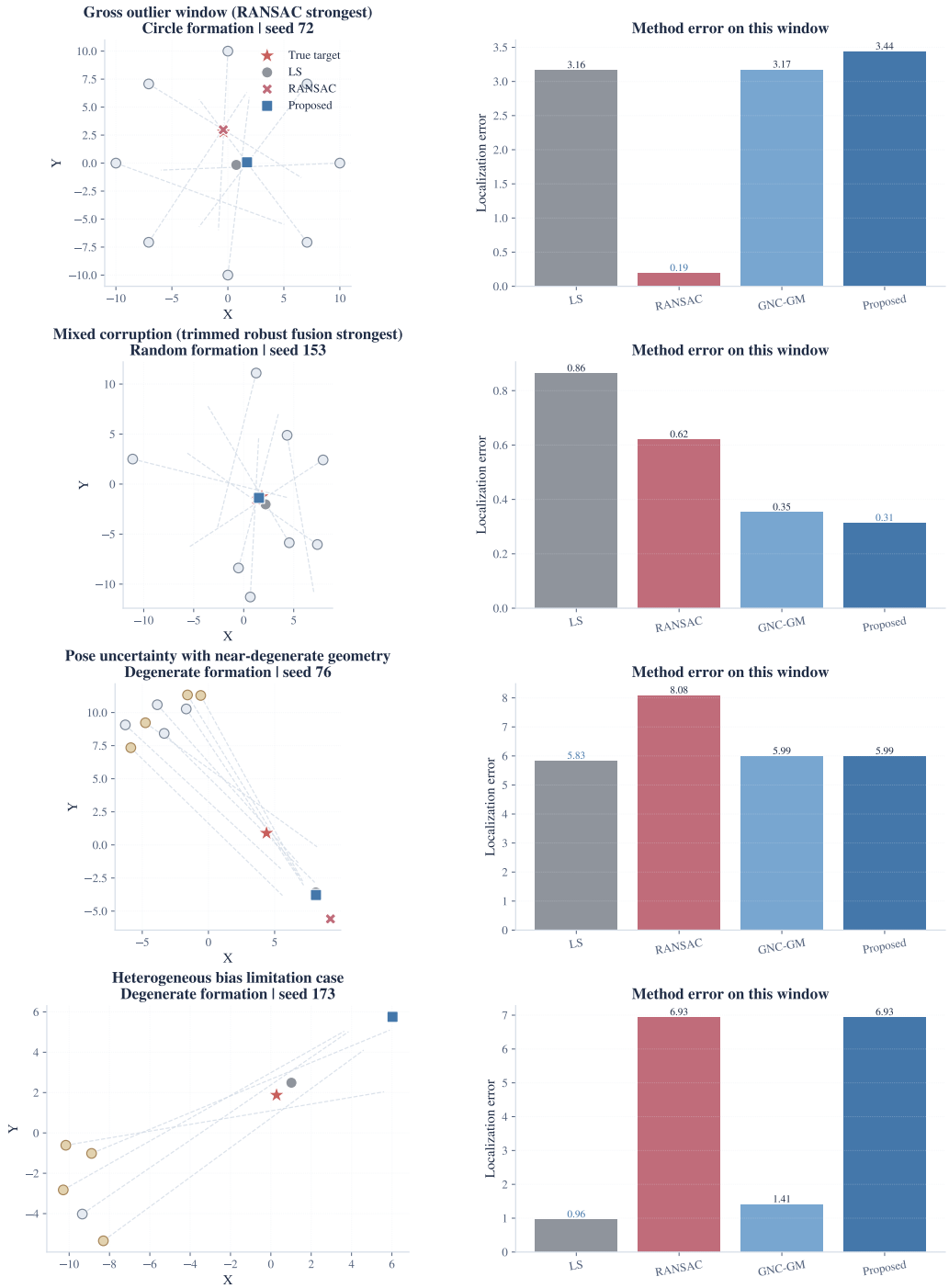

Figure 7. Four-case failure gallery. Row 1 shows a gross-outlier window in which RANSAC is strongest; Row 2 shows mixed corruption in which trimmed robust fusion is strongest; Row 3 shows a pose-uncertainty window with near-degenerate geometry; Row 4 shows a heterogeneous-bias limitation case.

The paired counts clarify the same point. The proposed always-on screening policy beats the residual-only baseline in 779 cases and loses in 603, with 903 ties. Against the all-sensor robust estimator, however, the same policy loses in 1370 cases and wins in only 661, which is why Stage 2 is not advertised as a replacement for full-set robust fusion. Adaptive screening behaves more conservatively: it improves over always-on screening in 604 cases, loses in 318, and ties in 1363 by refusing to screen when the current bearing set does not appear heterogeneous enough. The benefit map in Figure 9 shows the same behavior over the outlier-rate/budget plane: screening helps most when the budget is tight and measurement credibility is uneven.

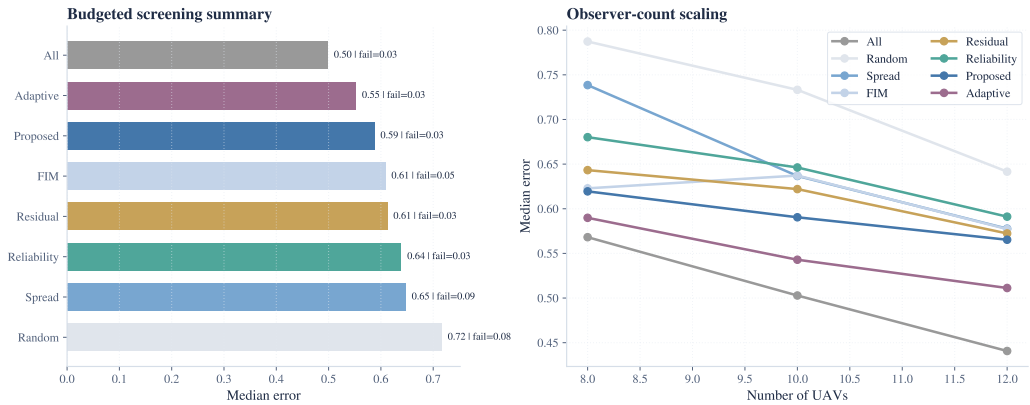

Figure 8. Fixed-budget screening benchmark over mixed and severe regimes. All-sensor robust fusion remains the unconstrained upper bound, while adaptive and geometry–reliability screening improve over simpler subset rules.

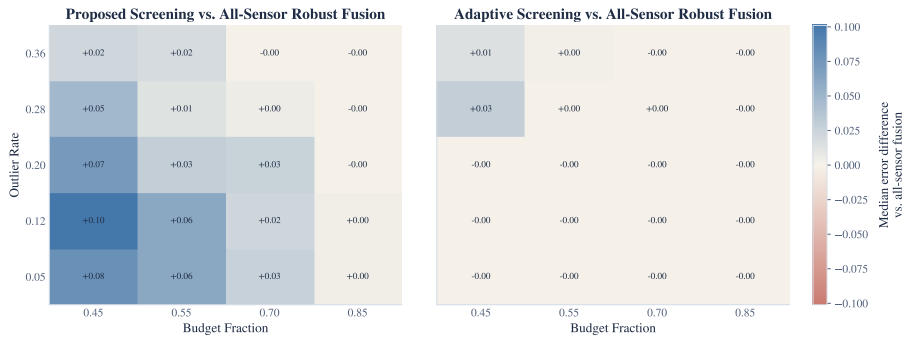

Figure 9. Selection benefit map over outlier rate and subset budget. The heat map reports the median error difference relative to all-sensor robust fusion; positive values indicate lower median error than the all-sensor baseline and show why screening should be activated selectively.

Figure 10 turns the conditional interpretation into two concrete case studies. In the first case, a severe perturbed 10-observer window with budget $k = 4$, the proposed subset policy attains error 0.2172, while residual-only and reliability-only selection both remain at 1.0680, the FIM-only subset remains at 1.0487, and the spread-only heuristic degrades to 3.9600. The all-sensor robust estimate is also worse at 1.0677, so this window shows why a budgeted subset can still be valuable when credibility heterogeneity is high. In the second case, a severe perturbed 12-observer window with budget $k = 6$, always-on screening would be the wrong decision: all-sensor robust fusion already achieves error 0.0368, while the always-on geometry–reliability policy degrades to 1.0637. The adaptive gate keeps the full set in that second case and therefore preserves the 0.0368 error. Together, these examples show why Stage 2 is best treated as a conditional engineering add-on for deadline- or uplink-limited cycles rather than as a universal replacement for full-set fusion.

Figure 11 addresses the second methodological concern: whether the proposed score needs every term or is dominated by one hand-crafted ingredient. Geometry-only screening

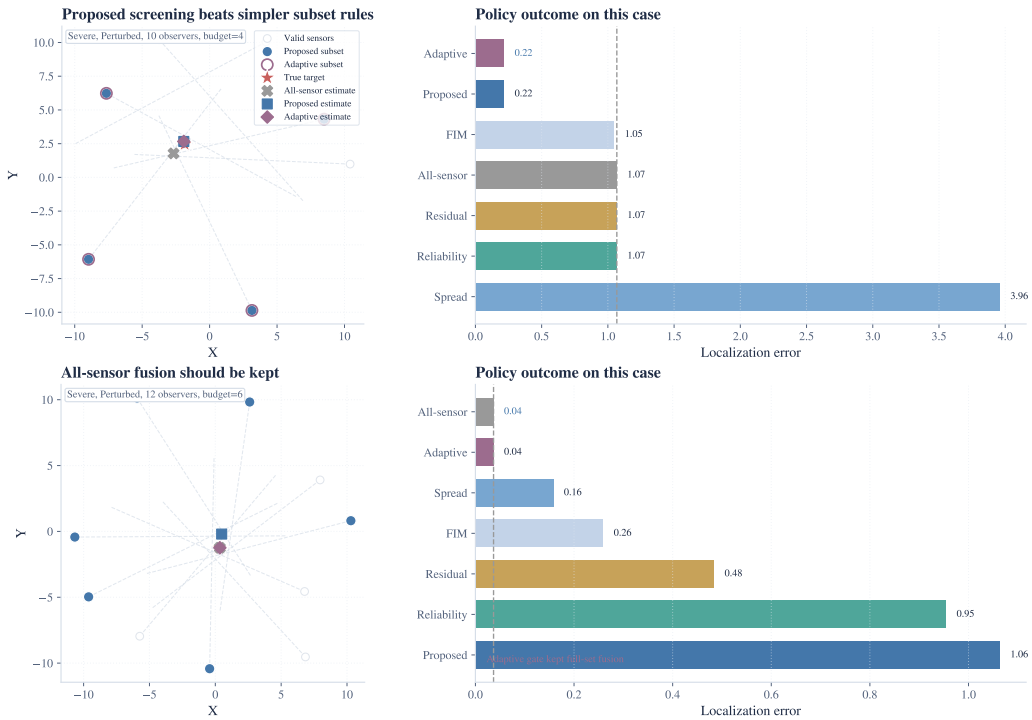

Figure 10. Two fixed-budget screening case studies. Top: a severe perturbed 10-observer window with budget $k = 4$, in which the proposed subset clearly outperforms simpler heuristics. Bottom: a severe perturbed 12-observer window with budget $k = 6$, in which the adaptive gate correctly keeps all-sensor fusion.

is the weakest screened variant, with overall median error 0.5857. Adding residual or reliability information narrows the gap, yielding medians of 0.5812 and 0.5641, respectively. The full score reaches 0.5790, which is not a universal optimum but remains in the same narrow range. Stage 2 should therefore be interpreted as an engineering compromise that integrates complementary cues without claiming dominance over every simpler screened rule.

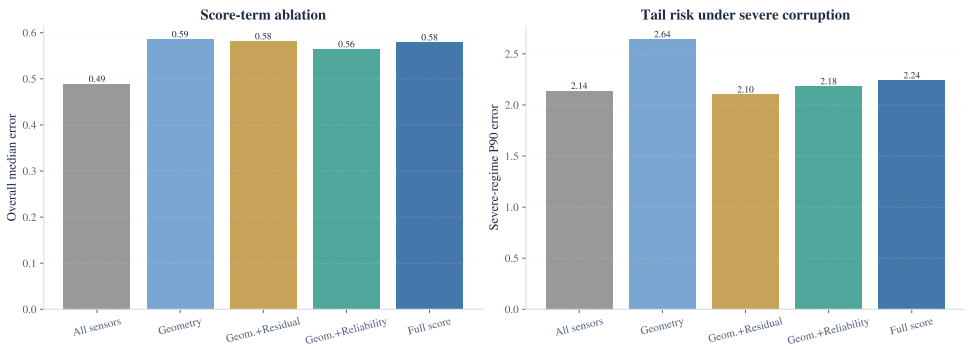

Figure 11. Score-term ablation for fixed-budget screening. Geometry-only screening is the weakest screened variant, while residual and reliability terms narrow the gap; the full score behaves as an engineering compromise rather than a universal optimum.

Figure 12 addresses the third methodological concern about the screening score: whether a nearby coefficient vector would change the conclusion completely. Across randomized mild, moderate, and strong perturbations of the four normalized coefficients, with the largest perturbation scale set to 16%, the default median error is 0.5586, the pooled 5th–95th percentile interval spans only 0.5479–0.5742, and the pooled median Jaccard overlap with the default selected subset remains 0.9685. These results do not make the score optimal, but they do show that the reported behavior is not a brittle artifact of one manually tuned coefficient vector.

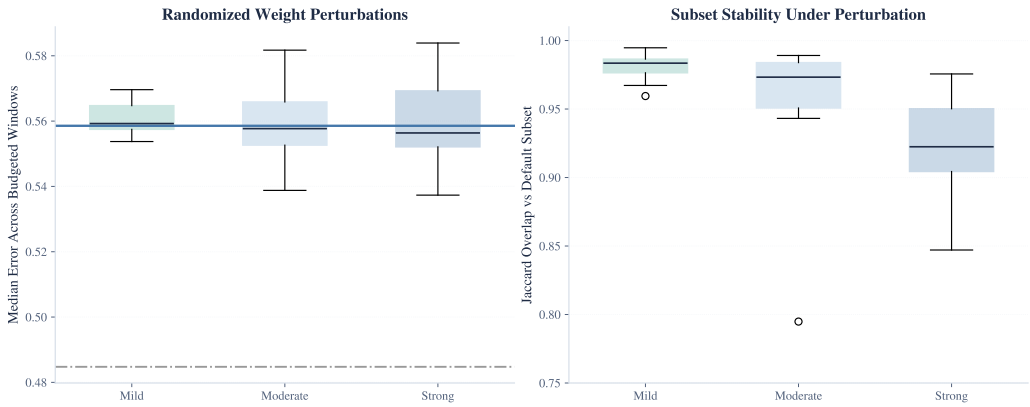

Figure 12. Sensitivity of the screening module under randomized mild, moderate, and strong coefficient perturbations. The budgeted median error remains close to the default setting, and the selected subsets stay strongly aligned with the default policy.

6.4. Pseudo-Physical Dynamic Replay

The pseudo-physical replay results show that the estimator is not only a static synthetic benchmark result. In the mild dynamic setting, the median error decreases from 0.1603 for least squares to 0.1343 for the proposed method, and the success rate at the 1.0 threshold reaches 1.0. In the harsh dynamic setting, the median error decreases from 0.4979 to 0.3523, while the 90th-percentile error drops from 1.5877 to 0.6505 (Table 9). The proposed method remains close to RANSAC in the mild setting and slightly better in the harsh setting, which is consistent with the main Monte Carlo story: consensus is important, but the additional trimmed refinement becomes more valuable when corruption sources combine.

Table 9. Pseudo-physical dynamic replay validation with delay, attitude jitter, and per-sensor bias. The proposed method is competitive with RANSAC in the mild setting and slightly better in the harsh setting while remaining clearly stronger than least squares and GNC-GM.

Regime	Method	Median	P90	Success@1.0	Cat.@5.0
Mild Dynamic	Least Squares	0.1603	0.6367	0.9722	0.0000
Mild Dynamic	GNC-GM	0.1496	0.3004	1.0000	0.0000
Mild Dynamic	RANSAC	0.1334	0.2635	1.0000	0.0000
Mild Dynamic	Proposed	0.1343	0.2681	1.0000	0.0000
Harsh Dynamic	Least Squares	0.4979	1.5877	0.7528	0.0028
Harsh Dynamic	GNC-GM	0.3647	0.7505	0.9694	0.0028
Harsh Dynamic	RANSAC	0.3559	0.6505	0.9778	0.0056
Harsh Dynamic	Proposed	0.3523	0.6505	0.9778	0.0056

6.5. Measured-Data Multi-View Replay Surrogate

The measured-data multi-view replay surrogate is built from surveyed observer positions, the RTK target trajectory, measured cross-camera synchronization, and real visibility masks from a public multi-view flight dataset [47]. For each synchronized and visible observer–target pair, a surrogate planar bearing is reconstructed as the azimuth of the line segment from the surveyed observer position to the RTK target point after projecting the 3D displacement into the local ground plane. Real geometry, visibility, and synchronization are therefore preserved by construction, while outlier, bias, and pose-noise corruption are injected only after this deterministic bearing reconstruction to probe front-end tail behavior under controlled stress. This layer therefore sits between a clean public replay and a fully synthetic benchmark.

Table 10 and Figure 13 show a clear trade-off. Least Squares attains the lowest nominal replay median at 32.1793 m, while GNC-GM attains the lowest disturbed replay median at

34.4460 m. The proposed front end is therefore not a center-error winner in this layer. Its advantage appears on the heavy tail. In nominal replay, the 95th-percentile error decreases from 241.7862 for least squares and 102.9909 for RANSAC to 78.8996, and the extreme-cue fraction above 80 m drops from 0.1323 and 0.0934 to 0.0506. In disturbed replay, the same pattern remains: the 95th percentile decreases from 287.8497 and 111.3714 to 105.2853, and the extreme-cue fraction decreases from 0.1219 and 0.1075 to 0.0968. These replay results show that the front end suppresses part of the extreme tail under measured asynchrony and visibility loss even when central replay accuracy remains comparable to or slightly behind stronger baselines.

Table 10. Measured-data multi-view replay surrogate built from surveyed observer locations, measured cross-camera synchronization, visibility masks, and RTK target trajectory in the public `drone-tracking-datasets` dataset. Least Squares gives the lowest nominal replay median, GNC-GM gives the lowest disturbed replay median, and the proposed front end yields the smallest upper-tail error in both replay regimes.

Regime	Method	Median (m)	P95 (m)	Ready@20 m	Extreme@80 m
Nominal Replay	Least Squares	32.1793	241.7862	0.2568	0.1323
Nominal Replay	GNC-GM	32.6326	218.9995	0.2607	0.1128
Nominal Replay	RANSAC	32.8595	102.9909	0.2335	0.0934
Nominal Replay	Proposed	34.6634	78.8996	0.2335	0.0506
Disturbed Replay	Least Squares	35.0541	287.8497	0.2509	0.1219
Disturbed Replay	GNC-GM	34.4460	283.9846	0.2688	0.1075
Disturbed Replay	RANSAC	36.3214	111.3714	0.2115	0.1075
Disturbed Replay	Proposed	36.8800	105.2853	0.2222	0.0968

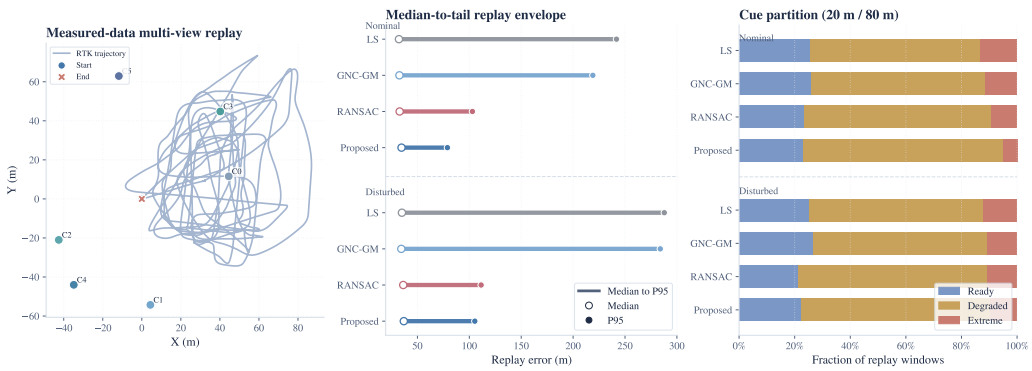

Figure 13. Measured-data multi-view replay surrogate built from surveyed observer locations, the RTK trajectory, and measured synchronization and visibility masks. Least Squares attains the lowest nominal replay median, GNC-GM attains the lowest disturbed replay median, and the proposed front end yields the smallest upper tail in both regimes.

6.6. Deadline-Aware Measured-Data Replay

The deadline-aware replay layer sharpens the operational question raised by the public replay: can the front end still deliver a usable cue when the current fusion cycle must ignore late or stale bearings? Figure 14 and Table 11 show that the answer remains yes in a bounded sense. The deadline filter removes more than half of the disturbed windows and reduces the average on-time bearing count to below two in the disturbed regime, so this is a materially harder setting than the plain measured-data replay.

Even under that filter, the proposed front end preserves the same qualitative strength. In nominal deadline replay, its median error is 33.9664 m, slightly behind least squares at 32.8007 m and GNC-GM at 33.7005 m, but its 95th-percentile error drops to 82.7327 m from 343.7387 m and 332.0415 m, and the extreme-cue fraction above 80 m falls to 0.0592 from 0.1316 and 0.1184. In disturbed deadline replay, the proposed method ties GNC-GM at

median 33.8389 m, improves over RANSAC at 36.5773 m, and contracts the 95th percentile to 86.0841 m from 121.6554 m for least squares and 104.9651 m for RANSAC. The same bounded conclusion therefore survives the additional systems filter: the method is not a universal center-error winner, but it reduces upper-tail cue failures when the front end is forced to discard bearings that arrive too late for the current cycle.

Table 11. Deadline-aware measured-data replay built on the same surveyed geometry, measured visibility masks, and synchronization tables as Table 10, but with an additional cycle-level arrival filter that discards stale or late bearings before localization. The nominal deadline layer retains 152/257 windows with 2.78 on-time bearings on average, while the disturbed layer retains 75/279 windows with 1.97 on-time bearings on average. The proposed front end remains comparable in median accuracy and yields the smallest upper tail in both deadline regimes.

Regime	Method	Median (m)	P95 (m)	Ready@20 m	Extreme@80 m
Nominal Deadline	Least Squares	32.8007	343.7387	0.2303	0.1316
Nominal Deadline	GNC-GM	33.7005	332.0415	0.2500	0.1184
Nominal Deadline	RANSAC	33.7950	139.6508	0.2500	0.1316
Nominal Deadline	Proposed	33.9664	82.7327	0.2566	0.0592
Disturbed Deadline	Least Squares	33.6946	121.6554	0.2933	0.0933
Disturbed Deadline	GNC-GM	33.8389	108.2326	0.2800	0.0667
Disturbed Deadline	RANSAC	36.5773	104.9651	0.2933	0.1200
Disturbed Deadline	Proposed	33.8389	86.0841	0.2800	0.0800

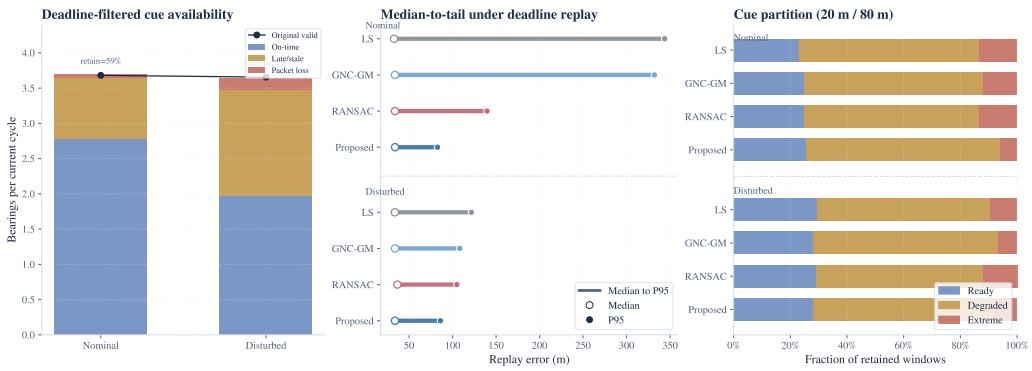

Figure 14. Deadline-aware measured-data replay. The left panel shows how many bearings remain usable after staleness, delay, and packet-loss filtering; the right panels show that the proposed front end keeps the smallest replay tail once only on-time bearings are fused.

6.7. Ablation and Sensitivity Trends

The ablation study helps explain why the method behaves like a front-end stack rather than like a single robustifier. In the compact mixed-regime ablation, removing the consensus seed produces the clearest degradation in stability: the success rate at $0.1R$ drops from 0.90 to 0.80 and the 90th-percentile error expands from 0.9997 to 2.1329, even though the median on some easier windows can remain small. By contrast, removing common-bias estimation changes the mixed-regime summary only slightly, which is consistent with the broader story that shared-bias correction matters mainly in more structured bias settings. Trimming and reweighting also show weaker effect in this particular compact ablation than in the larger main benchmark, indicating that their benefit is regime dependent rather than universal. Figure 15 shows that consensus seeding protects the initialization, while the other components provide additional robustness that becomes more visible in harsher or more structured corruption patterns.

The sensitivity sweeps tell a similarly nuanced story. As the outlier rate increases from 0.0 to 0.4, the proposed estimator stays below least squares throughout and remains below GNC-GM at the hardest outlier settings. Under increasing common bias, however, the

gap narrows and GNC-GM becomes competitive at the largest tested bias values. Under nominal-noise increase, the proposed estimator again remains clearly below least squares and is comparable to or slightly better than GNC-GM at higher noise levels. The front end is therefore strongest in outlier-rich and mixed-corruption settings rather than under every possible disturbance model.

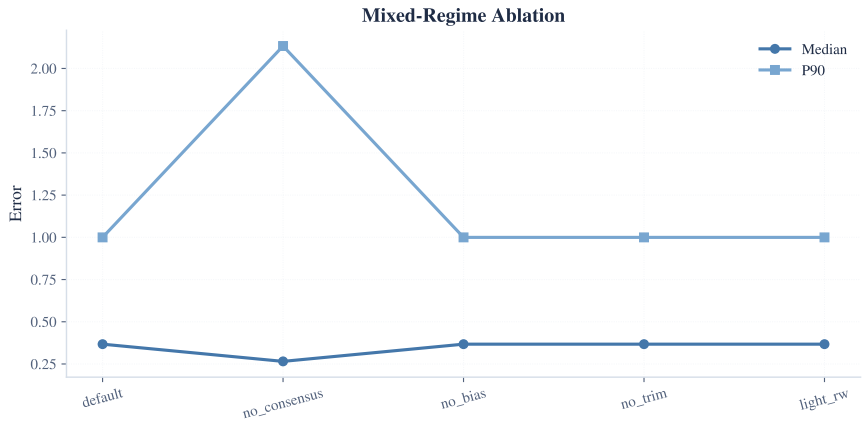

Figure 15. Mixed-regime ablation with component removal. The clearest degradation appears when the consensus seed is removed, while the other components produce smaller regime-dependent changes.

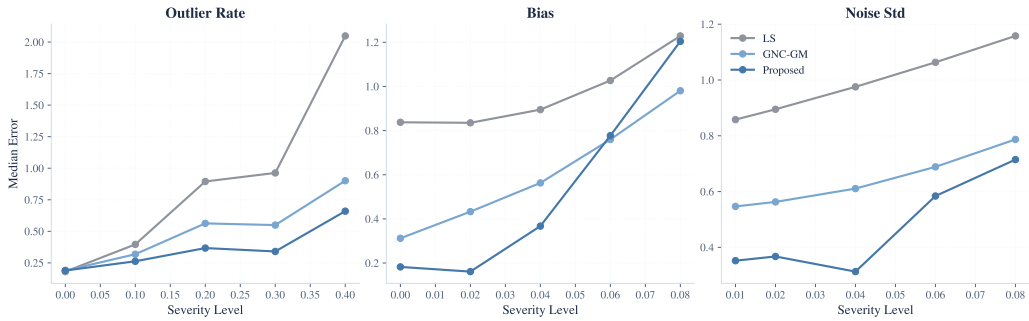

Figure 16. Sensitivity sweeps over outlier rate, bias magnitude, and nominal noise. The proposed estimator remains clearly stronger than least squares, while its advantage over GNC-GM depends on the corruption type.

6.8. Formation, Observability, and Sensor-Count Generalization

The formation study confirms that the proposed front end is not tied to one nominal geometry. Across circle, ellipse, perturbed, and random formations, the proposed estimator reduces the median error relative to least squares in every tested case (Appendix M, Table A19). The largest gain appears in the random formation, where the median drops from 2.1047 to 0.5768. The method is not uniformly best against every stronger baseline: GNC-GM remains competitive in some formation families, especially when the geometry is already favorable. The main advantage appears when corrupted bearings destabilize otherwise reasonable geometry.

Sensor-count scaling shows a similar pattern. The proposed estimator benefits substantially from moving from 4 to 8 or 10 observers, especially in random formations where the median decreases from 0.7221 at 4 observers to 0.4051 at 10 observers. The circle-formation scaling curve is also informative: the least-squares median is non-monotonic and can become worse at 12 observers under corruption, whereas the robust front end remains below 0.45 once eight or more observers are available. This suggests that additional corrupted measurements do not automatically help unless the fusion rule is robust.

The observability analysis adds an important nuance. Better isotropy still correlates with lower median error, but poor geometry alone does not explain the failures of least squares. Figure 19 shows that the proposed front end compresses the vertical spread of the error distribution over a wide isotropy range, meaning that it is addressing corrupted measurements rather than merely exploiting easy geometry.

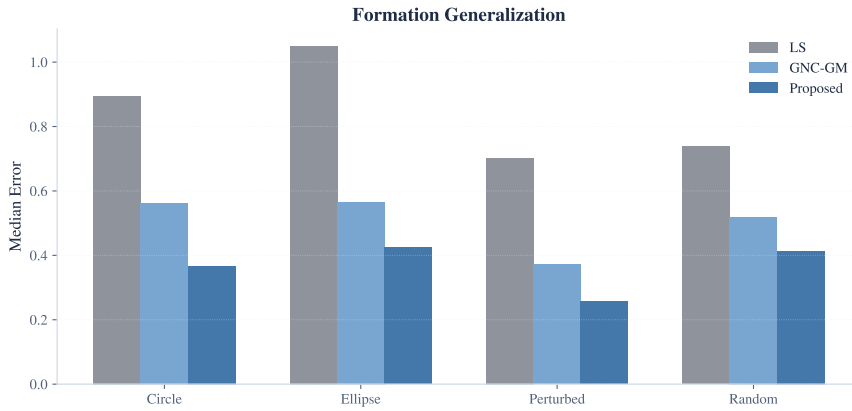

Figure 17. Formation generalization under mixed corruption. The proposed estimator stays below least squares across circle, ellipse, perturbed, and random formations.

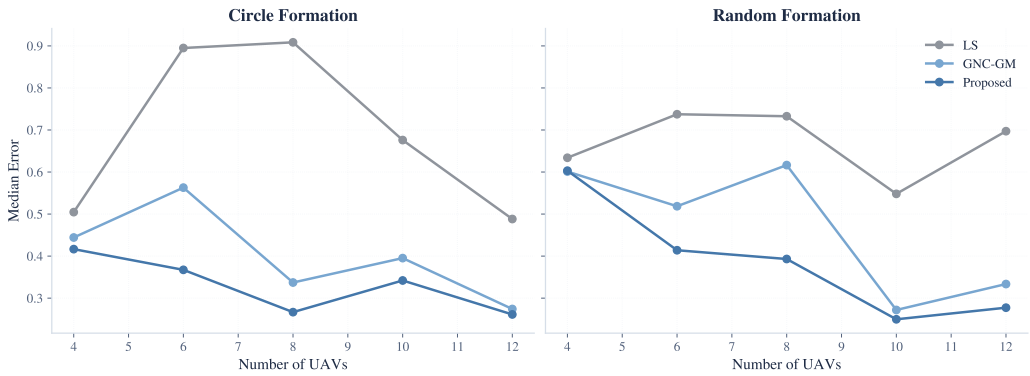

Figure 18. Sensor-count scaling for circle and random formations. Robust fusion becomes markedly more stable once eight or more observers are available.

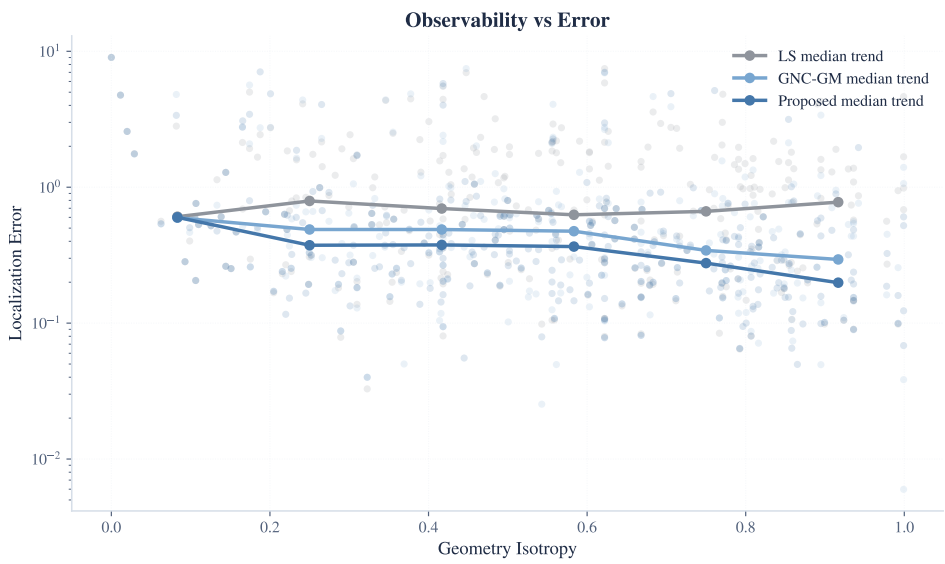

Figure 19. Error versus geometry isotropy. The proposed front end narrows the error spread even when the geometry is not especially favorable.

6.9. PyBullet Multi-Vehicle Replay

The PyBullet replay layer exposes the transfer gap more clearly than the purely synthetic benchmarks. Table 12 summarizes the replay configuration, including update rate, delay distribution, attitude-induced perturbation, and gust disturbance. Over 1103 replayed localization cycles, the overall median errors are 1.7469 for least squares, 1.6967 for GNC-GM, 2.0073 for RANSAC, and 2.0611 for the proposed estimator (Appendix M, Table A20). The proposed front end therefore does not dominate replay median accuracy or replay catastrophic-failure rate. Its main signal in this layer is tail-oriented: the mean error drops from 50.61 for least squares to 21.20, the overall 90th percentile drops from 16.69 to 13.11, and under disturbed replay the 90th percentile drops from 36.53 for least squares and 23.71 for RANSAC to 18.19 for the proposed method.

Table 12. PyBullet replay configuration for the semi-physical multi-UAV validation layer. Localization windows are sampled every three control steps, corresponding to a 16 Hz cue-update rate.

Parameter group	Setting	Value used in replay
Vehicle simulator	Engine	`gym-pybullet-drones`; `PYB_GND_DRAG_DW` physics
Timing	Rates	240 Hz simulation, 48 Hz control, 16 Hz localization update
Formation replay	Swarm layouts and counts	Circle and ellipse formations with 8 or 10 vehicles
Motion envelope	Horizon / orbit / altitude	7.0–7.5 s horizon, 6.0 s orbit period, 0.95 m base altitude with 0.04 m stagger
Bearing generation	FOV / yaw offset	$\pm 105^\circ$ half field of view with a -90° sensor-yaw offset relative to the body heading
Delay perturbation	Delay steps	Mean 0.45 / 0.60 control steps, standard deviation 0.22 / 0.28 in nominal / disturbed replay
Attitude-induced corruption	Gain pair	0.017 / 0.020 attitude gain and 0.0016 / 0.0020 yaw-rate gain in nominal / disturbed replay
Bias process	Shared bias / drift	0.0065 / 0.0090 shared bias with 0.0022 / 0.0030 sinusoidal drift
Random corruption	Missing / outlier rates	0.010 / 0.015 missing rate and 0.010 / 0.016 outlier rate in nominal / disturbed replay
Outlier magnitude	Outlier scale	0.08 / 0.12 radians in nominal / disturbed replay
Gust disturbance	Force std / trigger prob.	0.13 / 0.16 with 0.24 / 0.30 trigger probability in nominal / disturbed replay

Together with the measured-data replay surrogate, the PyBullet benchmark moves the evidence beyond a purely static algorithm study by incorporating multi-vehicle trajectories, timing variation, gust-like perturbation, and bias drift. This layer therefore provides a stronger bridge between synthetic geometry and UAV-like dynamics without being presented as field validation.

6.10. Cue-Quality Partition Under Replay

Median error alone does not fully explain whether a replayed localization window is useful to the rest of the mission stack. To make the application story more explicit, Figure 21 partitions replay windows into three cue classes using the same error thresholds already defined in Section 3: *handoff-ready* cues with error ≤ 1.0, *degraded* cues with error in $(1.0, 5.0]$, and *unusable* cues with error > 5.0. This is still a proxy rather than a full task-loop experiment, but it is a better description of replay transfer than a single mean or median.

The cue-quality partition clarifies why the PyBullet benchmark matters even though it is not a universal win. Over the overall replay pool, the proposed method produces a handoff-ready fraction of 0.3028 and an unusable fraction of 0.2584, which leaves it inside

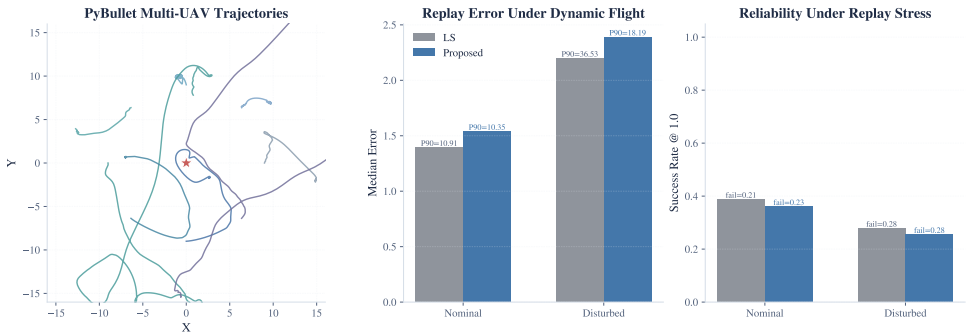

Figure 20. PyBullet multi-vehicle replay benchmark with delay, gust-like disturbance, and attitude-induced perturbation. The proposed front end does not dominate median replay accuracy, but it limits part of the upper tail in disturbed flight-like windows.

the same broad operating band as the stronger robust baselines rather than clearly ahead of them. Under disturbed replay, the same boundary remains: the handoff-ready fraction is 0.2546 and the unusable fraction is 0.2848. The replay layer therefore exposes a real transfer gap: static synthetic robustness does not automatically become better cue availability once delayed poses, attitude perturbation, and field-of-view effects are introduced together.

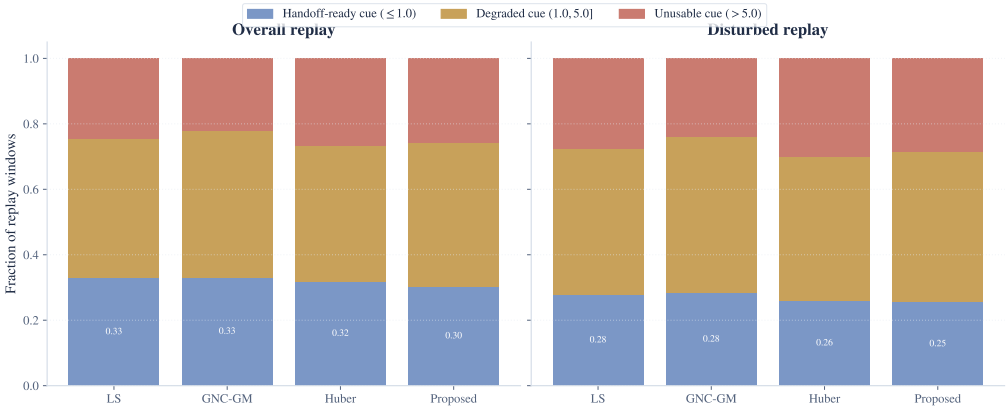

Figure 21. Cue-quality partition under PyBullet replay. Replay windows are grouped into handoff-ready, degraded, and unusable cues using the same normalized thresholds as the rest of the manuscript; the figure highlights a transfer gap rather than a universal replay win.

6.11. Downstream Tracking Proxy

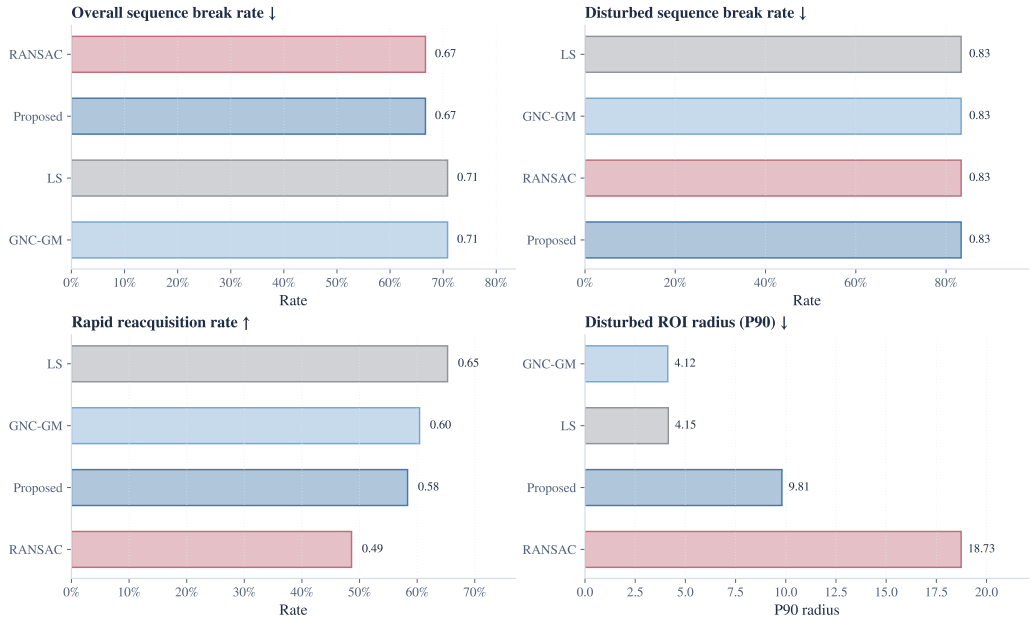

Figure 22. Tracking-proxy results produced by a minimal Kalman filter with chi-square gating. Once disturbed replay saturates, break-rate gains become limited, but the proposed front end still reduces the upper tail of tracking error relative to least squares and RANSAC.

To provide a lightweight downstream utility check, the replayed one-shot cues were also passed to a minimal Kalman filter with chi-square gating. The aim is to test whether a more reliable front-end cue remains beneficial once consumed by a simple tracker. Table 13 reports the numeric proxy metrics for all four front ends, and Figure 22 visualizes the same replay-to-tracker comparison. The overall replay-sequence break rate drops from 0.7083 for least squares and 0.7083 for GNC-GM to 0.6667 for both RANSAC and the proposed front end. In overall replay, the tracking-RMSE P90 is 16.4994 for the proposed front end and 16.6907 for GNC-GM, both well below least squares at 28.8386 and RANSAC at 34.3885. Under disturbed replay, all four methods share the same break rate of 0.8333; GNC-GM attains the smallest tracking-error P90 at 12.9715, the proposed front end follows at 13.6756, and both remain far below least squares and RANSAC.

The additional operational proxies are reported even though they are mixed. In disturbed replay, the proposed method reaches a rapid reacquisition rate of 0.5833, compared with 0.6528 for least squares, 0.6042 for GNC-GM, and 0.4861 for RANSAC, while the disturbed ROI-radius P90 is 9.8080 compared with 4.1488 for least squares, 4.1218 for GNC-GM, and 18.7335 for RANSAC. These numbers show that the present front end reduces part of the downstream heavy tail and can shorten recovery after some corrupted cues, but it does not dominate every tracking proxy.

This proxy remains modest. RANSAC matches the overall break-rate reduction, and GNC-GM is slightly better than the proposed method on some disturbed tracking proxies. The operational signal is therefore narrow: the proposed front end keeps the best overall break rate, nearly matches GNC-GM on the overall tracking tail, and remains well below least squares and RANSAC on disturbed tracking-error P90 even though disturbed break rates are saturated.

6.12. Runtime and Reproducibility Implications

The runtime results show that the robustness gain does not require an obviously impractical computational budget, although the method is clearly slower than simple least-squares fitting.

Table 13. Downstream tracking proxy on PyBullet replay using a minimal Kalman filter with chi-square gating. Sequence break means at least one three-miss gating failure within a replay sequence. All four front ends are reported because replay-level robustness ordering does not fully predict downstream tail behavior.

Method	Overall Break Rate	Overall P90 RMSE	Disturbed Break Rate	Disturbed P90 RMSE	Disturbed Rapid Reacq.	Disturbed ROI P90
Least Squares	0.7083	28.8386	0.8333	29.8760	0.6528	4.1488
GNC-GM	0.7083	16.6907	0.8333	12.9715	0.6042	4.1218
RANSAC	0.6667	34.3885	0.8333	36.5509	0.4861	18.7335
Proposed	0.6667	16.4994	0.8333	13.6756	0.5833	9.8080

The reported desktop measurements were obtained on an Apple M1 Pro CPU with 10 logical cores running macOS 15.7.2 and Python 3.11.11. Under this setup, the proposed front end has a median runtime of 23.82 ms, compared with 1.40 ms for least squares, 1.44 ms for Huber IRLS, 2.61 ms for RANSAC, and 9.98 ms for GNC-GM. Figure 23 separates method-level cycle latency from stage-wise scaling with observer count. The robust Stage 1 core has median runtime 23.65 ms, while Stage 2 screening adds 1.86 ms and the adaptive gate adds 1.58 ms when invoked. A 23.82 ms cycle corresponds to roughly 42 Hz serial execution, which means a 10 Hz front-end fusion cadence is feasible on the reported desktop CPU with comfortable margin. This statement applies only to the reported desktop setting; deployment on an onboard companion computer would still require C++ reimplementation or explicit parallelization.

Equally important is the reproducibility structure. All manuscript figures are generated from frozen JSON outputs, and the experiment runners for the main benchmark, the screening study, the measured-data replay surrogate, the pseudo-physical replay, and the PyBullet replay are archived in the repository. This matters because the paper is intended to be reused as a benchmark baseline. The benchmark and appendix materials summarized in Appendices L and M therefore form part of the contribution, not only supplementary packaging.

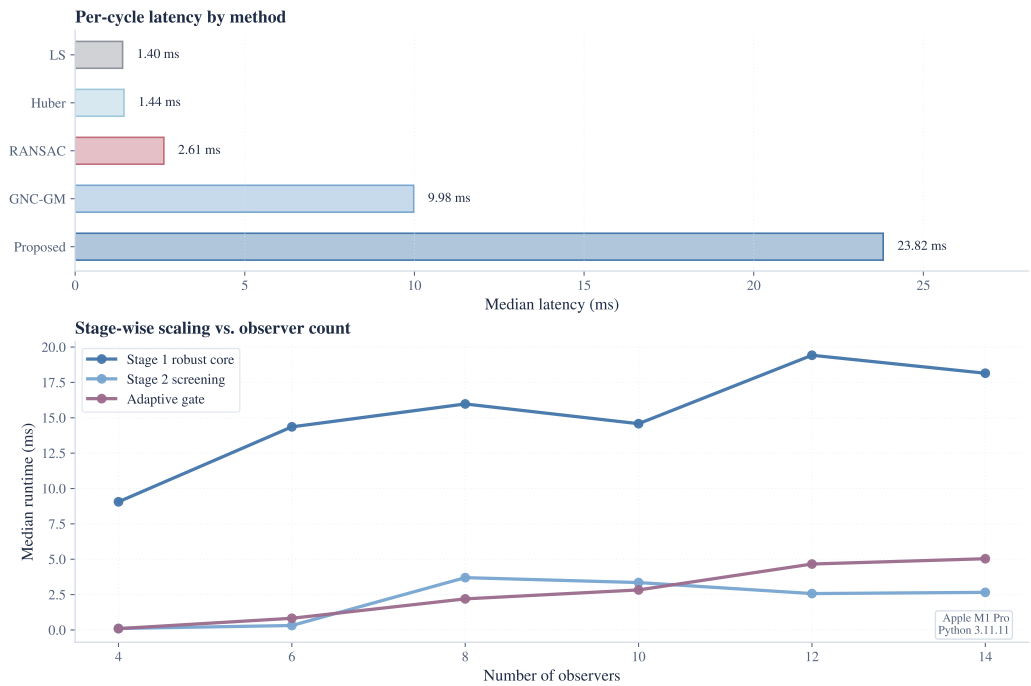

Figure 23. Runtime profile on the reported desktop CPU. The upper panel reports per-cycle median latency by method, and the lower panel shows stage-wise scaling with observer count. The proposed front end remains slower than least squares, Huber, and RANSAC, but still fits desktop cycle-level validation.

7. Discussion

7.1. Operating Regimes and Tail Behavior

The strongest evidence comes from outlier-rich, mixed-corruption, and disturbed-replay settings. In those windows, the main benefit of the proposed front end is not universal center-error dominance but heavy-tail suppression. A tracker, handoff module, or downstream search procedure can often tolerate moderate noise more easily than catastrophic cue displacement. The proposed estimator therefore adds value primarily by preventing a small number of corrupted cycles from dominating the mission outcome.

The baseline comparison supports the same interpretation. Least Squares is clearly too brittle once corruption becomes non-negligible, while GNC-GM and RANSAC already provide strong robust references. Against those stronger baselines, the proposed estimator remains comparable in center accuracy and stronger in several tail-oriented metrics under mixed corruption, heterogeneous bias, and disturbed replay. The same conditional reading applies to Stage 2: budgeted screening is useful when credibility heterogeneity is high or when only part of the current bearing set can be fused, whereas all-sensor robust fusion remains the correct default when the full set is available and coherent.

7.2. Implications for Passive Multi-UAV Sensing

Taken together, Figures 2, 13, 14, 21, and 22 support a concrete system interpretation. The method improves the reliability of the current-cycle target cue before it is consumed by downstream modules such as tracking, handoff, or replanning. The measured-data multi-view replay surrogate strengthens this claim by exposing the front end to real observer geometry, visibility loss, and synchronization offsets; the deadline-aware replay layer then adds a cycle-level arrival filter that removes stale or late measurements before fusion; and the PyBullet replay layer adds multi-vehicle motion, delay, and attitude-induced perturbation. The downstream Kalman proxy then shows that reduced localization tail risk can translate into smaller tracking-error tails even when break-rate improvements are limited.

This validation chain also clarifies what the present study contributes to a UAV sensing pipeline. The paper isolates a front-end failure mode that frequently appears before longer-horizon modules can help: a partially corrupted set of bearings must still be fused into a usable cue inside the current cycle. The resulting contribution is a current-cycle robust-localization block with explicit interfaces to upstream poses and bearings and to downstream tracking, handoff, or replanning modules.

7.3. Limitations and Next Steps

Several limitations remain. The current formulation is planar and centralized, and the observer poses are assumed to be supplied by an external navigation module rather than estimated jointly with the target. Although pose uncertainty is injected into the benchmark, the estimator does not yet use covariance-aware weighting or a joint uncertainty propagation model inside the refinement step. The screening score is still a fixed engineering surrogate rather than a learned or theoretically optimal budgeted selector. The validation chain now includes measured-data replay, deadline-aware measured-data replay, pseudo-physical replay, and PyBullet replay, but it still stops short of SITL, HITL, or onboard field validation.

These limitations point directly to the next development steps. The most immediate extension is to replace the current pose-agnostic weighting with covariance-aware refinement so that navigation uncertainty is handled inside the estimator rather than only in the benchmark. A second extension is to move from the current planar formulation to full three-dimensional bearing fusion with altitude variation and camera or antenna mounting

geometry. A third extension is system-level integration: coupling the present front end to richer trackers, explicit communication scheduling, and onboard companion-computer implementations. The reported desktop timings show that the current Python prototype is already usable for algorithm studies, but practical airborne deployment will likely require C++ reimplementation or parallelization on the onboard companion computer.

8. Conclusions

This paper formulates the contribution as a corruption-aware one-shot bearing-only localization problem and presents a robust front end for that problem. The consensus-assisted estimator suppresses catastrophic one-shot failures caused by bad bearings, while the optional adaptive screening add-on improves over simpler fixed-budget subset policies without replacing all-sensor fusion when the full measurement set is available. Across 3000 Monte Carlo cases, 536 measured-data multi-view replay windows, 227 deadline-filtered replay windows, and 720 pseudo-physical dynamic replay cases, the method consistently improves over least squares. Against the strongest robust baselines, the evidence is more bounded: in measured-data replay, Least Squares and GNC-GM retain the best medians in the nominal and disturbed subsets, while in deadline-aware replay the strongest baselines still set several center-error values, but the proposed front end repeatedly produces the smallest upper tail; in PyBullet replay, median accuracy remains comparable while tail-oriented tracking metrics improve in disturbed windows.

The main value of the study is therefore a reproducible tail-risk suppression baseline for passive bearing-only localization under outliers, mixed corruption, observer-pose uncertainty, and heterogeneous bias. Future work should add covariance-aware refinement, extend the formulation to full three-dimensional localization, connect the front-end output to richer downstream task metrics, and validate the pipeline through SITL, HITL, or onboard experiments.

Appendix I Representative Positioning Against Nearby Literature

Table A14. Positioning of the present manuscript relative to representative recent literature.

Reference	Primary focus	Corrupted-bearing front end	Fixed-budget screening	Validation layer
[3]	UAV localization survey and taxonomy	No	No	Survey paper
[16]	Passive positioning and formation electromagnetic compatibility for UAV swarms	Not explicit	No	Simulation-focused
[5]	Bearing-only passive localization with formation adjustment under electromagnetic silence	Partial geometry/bias discussion	No	Simulation-focused
[24]	Dual-UAV cooperative bearing-only localization with particle filtering	Yes, through filtering	No	Algorithm simulation
This work	One-shot robust fusion under outliers, bias, missing data, and pose uncertainty	Yes	Yes	Synthetic, pseudo-physical replay, and PyBullet replay

Appendix J Threats to Validity

The most important construct-validity threat is that corrupted bearings are represented by parameterized stochastic models rather than by an onboard sensor stack. The pseudo-physical and PyBullet replay layers reduce this gap, but they do not remove it. The most

important internal-validity threat is that the screening module remains an engineered score with fixed coefficients, although the randomized coefficient-perturbation study shows that its behavior is not tied to one brittle weight vector. The most important external-validity threat is that the study remains planar and centralized. These limits define the current scope and motivate future extensions toward SITL, HITL, field-log replay, and full three-dimensional localization.

Appendix K Sketch of a 3D Extension

The same front-end structure extends naturally to three-dimensional passive localization if each observer pose is represented as $s_i = [x_i, y_i, z_i]^T$ and the target state is represented as $p = [x, y, z]^T$. A compact measurement model is

$$\mathbf{z}_i = \begin{bmatrix} \psi_i \\ \varphi_i \end{bmatrix} = \begin{bmatrix} \operatorname{atan2}(y - y_i, x - x_i) \\ \operatorname{atan2}\Big(z - z_i, \sqrt{(x - x_i)^2 + (y - y_i)^2}\Big) \end{bmatrix} \mathbf{b}_i + \boldsymbol{\epsilon}_i + \mathbf{o}_i, \tag{A18}$$

where ψ_i and φ_i denote azimuth and elevation, respectively. The 3D front end would keep the same two-stage logic: consensus-style seeding, trimmed robust refinement, and optional budget-aware screening, but the Jacobian and FIM terms would become 3×3 objects and the observability score would need to reflect both horizontal and vertical geometry. A distributed V2V version would then decide whether to transmit full angle packets, compressed sufficient statistics, or only screened subsets.

Appendix L Benchmark Matrix and Hyperparameters

Table A15. Benchmark matrix used in the manuscript.

Benchmark layer	Cases	Formations / counts	Main perturbations	Main purpose
Main Monte Carlo	3000	Circle, random, and degenerate; 8 observers	Outliers, mixed corruption, pose noise, heterogeneous bias	Establish the core robust-localization story
Fixed-budget screening	2285	Circle, perturbed, random, and degenerate; 8/10/12 observers	Mixed and severe corruption under subset budgets	Test whether screening helps when not all bearings can be fused
Measured-data multi-view replay surrogate	536	Surveyed multi-view observers; real target trajectory	Measured visibility loss, measured asynchrony, pose noise, bias, outliers	Add a measured-data replay layer between synthetic and replay evidence
Deadline-aware measured-data replay	227	Same surveyed observer network; synchronized replay windows	Measured visibility and synchronization plus sampled link delay, packet loss, and staleness filtering	Test whether the front end remains usable when only on-time bearings survive the current cycle
Pseudo-physical replay	720	Three formations; three swarm sizes	Delay, jitter, missing data, per-sensor bias, outliers	Add flight-like disturbances without a full vehicle simulator
PyBullet multi-vehicle replay	1103	Circle and ellipse; 8/10 observers	Attitude disturbance, gust-like perturbation, delay, bias drift, missing data	Evaluate the estimator on multi-vehicle trajectories and asynchronous logs
Generalization / efficiency sweeps	Multiple sweeps	Four formations; 4–12 observers; five severity levels	Outlier-rate, bias, noise, observability, timing cost	Characterize scaling, sensitivity, and computational burden

Table A16. Key algorithmic hyperparameters used by the robust estimator and the screening module.

Setting	Default	Role	Stress-tested in
`trim_ratio`	0.15	Removes the largest residuals before reweighting, capped to keep at least three active bearings	Ablation study
`reweight_iters`	3	Number of outer robust-weight updates	Ablation study
`huber_delta`	0.08	Residual scale for initial reliability estimation	Sensitivity and screening benchmarks
`tukey_c`	0.18	Tail cutoff for aggressive down-weighting during refinement	Ablation study
`lm_damping`	10^{-3}	Damping for Gauss–Newton / LM updates	All benchmarks
`max_step_norm`	3.0	Prevents unstable jumps from poor initial hypotheses	Dynamic and PyBullet replay
`ransac_iterations`	72	Consensus fallback search depth	Main Monte Carlo and replay
`ransac_thr`	0.12	Bearing residual threshold for the fallback hypothesis	Main Monte Carlo and replay
Adaptive gate	MAD $<$ 0.05 and $\sigma_w < 0.12$	Keeps all-sensor fusion in cleaner windows	Fixed-budget screening
Candidate pool size	$\max(k + 2, 8)$	Limits exhaustive search to a tractable top-reliability pool	Fixed-budget screening

Appendix M Supplementary Result Tables

Table A17. Selected points from the sensitivity sweeps.

Sweep	Level	LS	GNC-GM	Proposed
Outlier rate	0.0	0.1808	0.1893	0.1893
Outlier rate	0.2	0.8950	0.5629	0.3673
Outlier rate	0.4	2.0484	0.9010	0.6599
Bias	0.00	0.8377	0.3121	0.1824
Bias	0.04	0.8950	0.5629	0.3673
Bias	0.08	1.2289	0.9805	1.2045
Noise std	0.01	0.8580	0.5467	0.3522
Noise std	0.04	0.9754	0.6107	0.3130
Noise std	0.08	1.1580	0.7869	0.7146

Table A18. Sensor-count scaling medians for circle and random formations.

Formation	Observers	LS	GNC-GM	Proposed
Circle	4	0.5046	0.4443	0.4166
Circle	6	0.8950	0.5629	0.3673
Circle	8	0.9085	0.3372	0.2667
Circle	10	0.6760	0.3952	0.3420
Circle	12	0.4882	0.2742	0.2613
Random	4	0.6340	0.6012	0.6032
Random	6	0.7375	0.5186	0.4140
Random	8	0.7327	0.6165	0.3931
Random	10	0.5481	0.2719	0.2497
Random	12	0.6972	0.3335	0.2773

Table A19. Formation-generalization medians in the mixed-corruption study.

Formation	LS	GNC-GM	Proposed
Circle	0.8950	0.5629	0.3673
Ellipse	1.0501	0.5661	0.4244
Perturbed	0.7024	0.3723	0.2574
Random	0.7375	0.5186	0.4140

Table A20. PyBullet multi-vehicle replay summary. Median error remains close to the stronger robust baselines, while several tail-oriented indicators improve under disturbed replay.

Regime	Method	Median	P90	Success@1.0	Cat.@5.0
Overall	Least Squares	1.7469	16.6860	0.3282	0.2448
Overall	GNC-GM	1.6967	12.7876	0.3309	0.2221
Overall	RANSAC	2.0073	15.0330	0.3200	0.2593
Overall	Proposed	2.0611	13.1131	0.3028	0.2584
Nominal	Least Squares	1.3930	10.9102	0.3874	0.2075
Nominal	Proposed	1.5377	10.3484	0.3597	0.2273
Disturbed	Least Squares	2.1975	36.5283	0.2781	0.2764
Disturbed	Proposed	2.3891	18.1859	0.2546	0.2848

Table A21. Expanded paired-significance summary from the main 3000-case synthetic benchmark. Positive median improvement means lower error for the proposed front-end family.

Regime	Comparison	Wins–Losses–Ties	Wilcoxon p	Median improvement
Outlier	Proposed vs LS	521–70–9	2.35×10^{-77}	0.7637
Outlier	Proposed vs GNC-GM	385–134–81	6.16×10^{-33}	0.0953
Outlier	Proposed vs RANSAC	93–86–421	0.0357	0.0000
Mixed	Proposed vs LS	411–169–20	2.39×10^{-42}	0.3258
Mixed	Proposed vs GNC-GM	293–161–146	2.32×10^{-14}	0.0000
Mixed	Proposed vs RANSAC	78–107–415	0.9992	0.0000
Pose uncertainty	Proposed vs LS	394–189–17	1.21×10^{-32}	0.0680
Pose uncertainty	Proposed vs GNC-GM	262–195–143	5.06×10^{-6}	0.0000
Pose uncertainty	Proposed vs RANSAC	127–114–359	0.1933	0.0000
Heterogeneous bias	Proposed vs LS	338–240–22	2.33×10^{-13}	0.0025
Heterogeneous bias	Proposed vs GNC-GM	184–196–220	0.2640	0.0000
Heterogeneous bias	Proposed vs RANSAC	136–150–314	0.8396	0.0000

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